

# **Module-3**

## **Special Theory of Relativity:**

Introduction, Inertial frame of reference, Galilean transformations, Postulates, Lorentz transformations and its conclusions, Length contraction, time dilation, velocity addition, Mass change, Einstein's mass energy relation.

- Michelson – Morley Experiment
- Postulates of Special Theory of Relativity
- Lorentz Transformations
- Length Contraction
- Time Dilation
- Experimental Verification of Time Dilation
- Relativity of Velocity
- Relativity of Mass
- Mass – Energy Equivalence

# The Science of Physics is often divided into Classical Physics and Modern Physics.

**Classical Physics** is a model of the macroscopic world around us. All the laws of classical physics were known by the end of the 19<sup>th</sup> century. Classical physics works well describing and predicting almost all everyday phenomena. The known properties of matter at the end of the 19th century were mass and charge. The smallest constituents were atoms.

The known interactions were gravity, modeled by **Newton's law of gravitation**, electromagnetic interactions, modeled by **Maxwell's equations**, and **contact force** arising from the requirement that "atoms need their space".

Consequences of the interactions were described by **Newton's laws of motion**, which predict how matter behaves when acted on by forces. Statistical physics and thermodynamics were developed for describing systems with a large number of degrees of freedom.

**Modern Physics** is the physics of the 20<sup>th</sup> century. The main building blocks, the **theory of relativity** and **quantum mechanics**, were developed early in that century.

## THE TWO PROBLEMS WITH CLASSICAL PHYSICS THAT BECAME APPARENT AT THE END OF

### THE 19TH CENTURY WERE:-

**1. The ultraviolet catastrophe:** According to classical physics, the energy emitted by a blackbody should increase indefinitely as the frequency of the radiation increases.

This means that at high frequencies, the radiation should have infinite energy, which is obviously not observed. This was known as the "ultraviolet catastrophe" because it predicted an infinite amount of energy in the ultraviolet range of the electromagnetic spectrum.

**2. The photoelectric effect:** Another problem with classical physics was that it could not explain the photoelectric effect, which is the observation that electrons are emitted from a metal surface when it is exposed to light of a certain frequency. According to classical physics, the energy of the electrons emitted should increase as the intensity of the light increases, but this is not what is observed. Instead, the energy of the electrons depends only on the frequency of the light, which is not predicted by classical physics.

These problems led to the development of quantum mechanics, which is a new framework for describing the behavior of matter and energy on a microscopic

## Classical Physics

### Matter

Atoms

Stuff made from atoms

### Interactions

Fields and contact forces

Newton's law of gravitation

Maxwell's equations (EM)

What does matter do as a result of these interaction?

Newton's laws of motion:  $\mathbf{F} = m\mathbf{a}$ ,  $\mathbf{F}_{12} = -\mathbf{F}_{21}$

## Modern Physics

Building blocks:

**theory of relativity:**

Removes inconsistencies in the classical theory.  
Describes the behavior of matter at **high energies**  
and **high speeds**.

Classical physics is easily derived as a non-relativistic ( $v \ll c$ ) limit.

**quantum mechanics:**

Removes disagreements between theory and  
measurements.  
Describes the behavior of **microscopic particles**.

Predictions of classical physics are **not** easily derived by making a transition to  
the macroscopic world.

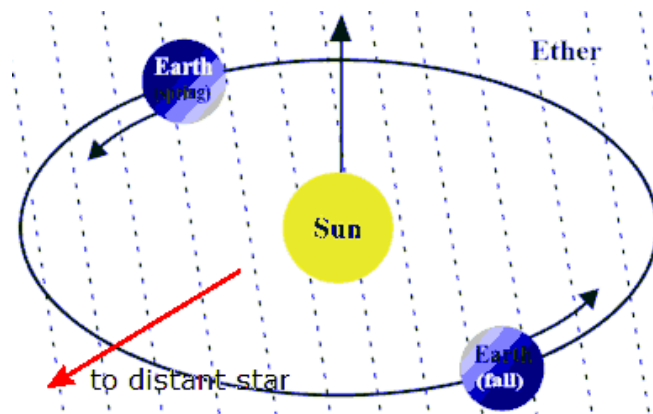
**It all started with light!**

### Example:

You walk from the back to the front of the airplane with a speed of 1 m/s relative to the plane. The airplane is flying with a speed of 200 m/s in the forward direction with respect to the surface of Earth. Your speed with respect to the surface of Earth is 201 m/s.

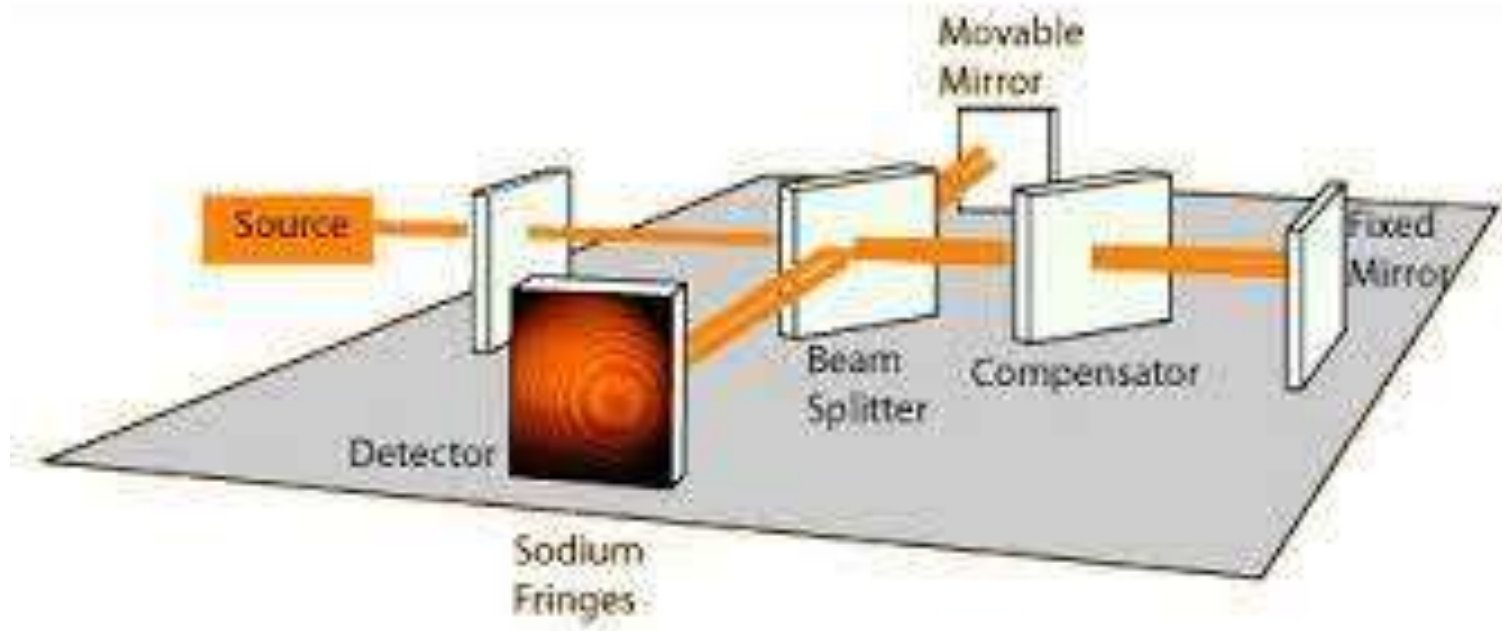
You send a pulse of light from the back to the front of the airplane. Maxwell's equations predict the speed of that pulse to be  $c$  with respect to the airplane and with respect to the surface of the earth. What gives?

The first inclination of physicists was to assume that even though Maxwell did not specify a reference frame in which light moved with speed  $c$ , such a unique reference frame surely had to exist. It was called the ether, some type of substance that permeated all space, and through which light waves propagated with speed  $c$ . Experiments were looking for this ether, but never found it.



A famous experiment, called the **Michelson-Morley experiment**, used an interferometer to make high-precision measurements of the speed of light emitted by a distant star with respect to Earth. Measurements were made at different times of the year, when Earth was moving towards or away from the star in its orbit around the sun.

The Michelson-Morley experiment found no dependence of the speed of light with respect to Earth on the time of the measurement and is often cited as the definite experiment disproving the existence of the ether.



Einstein special theory of relativity resolves the conflict between Newton's laws of motion and Maxwell's equations.

Einstein proposed two postulates.

- I. The laws of nature are the same in all inertial reference frames.
- II. In vacuum, light propagates with respect to any inertial frame and in all directions with the universal speed  $c$ . This speed is a constant of nature.



# Introduction:

The theory of relativity was successfully described by Einstein in 1905. It was related to the relative motion of the objects or we can say it deals with the motion of object or observer moving with relative velocities w.r.t. each other.

This theory has two parts:-

## **1. Special Theory of Relativity**

(What is Spatial in it:- Object is moving constant velocity and the velocity of object will be comparable to the velocity of Light)

Special theory of relativity was formulated by A. Einstein in 1905. In this theory we deal with the relative motion of the objects, which are moving with constant velocity or are at rest w.r.t. one another. **The main aim of this theory was to solve the problem of absolute motion of bodies.**

## **2. General Theory of Relativity**

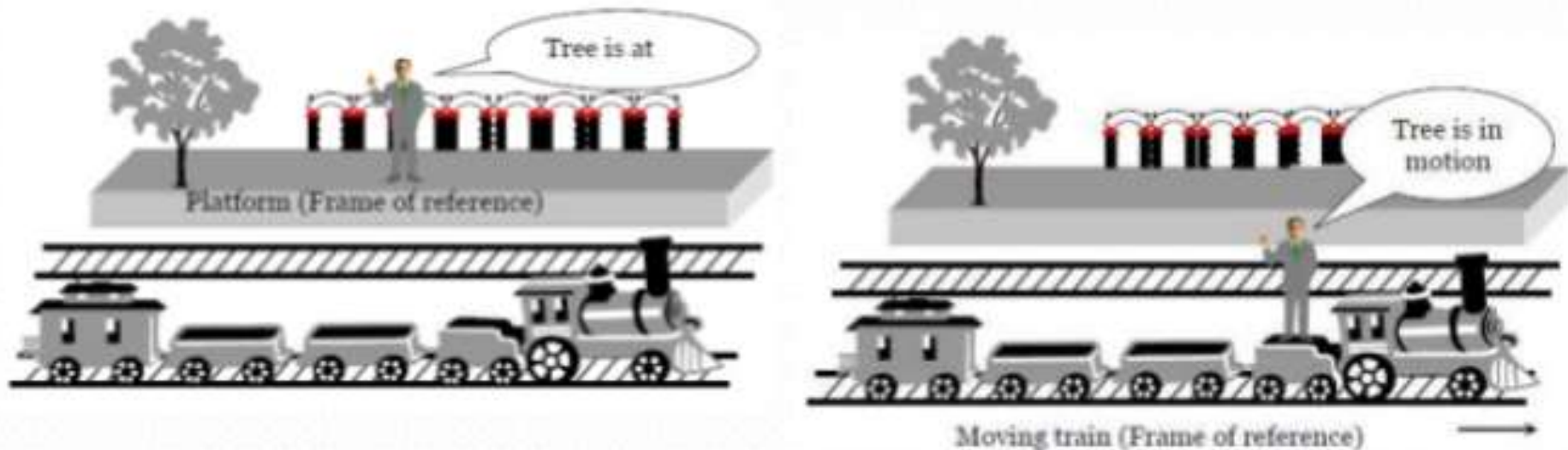
But Einstein found that there is no absolute motion in free space.

In general theory of relativity (1915) we deal with the problem that involves frame of reference accelerated w.r.t. one another.

## Frame of Reference:

Any system relative to which the motion of an object can be described is called a **frame of reference**. It is selected in such a way that the laws of nature become fundamentally simpler in that frame of reference.

It is a system to which a set of coordinates are attached and with reference to which observer describes any event

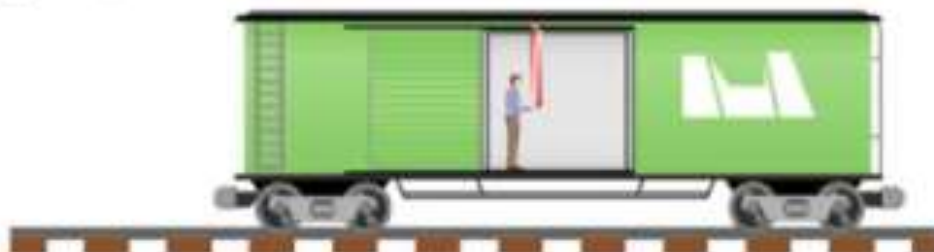


A passenger standing on platform observes that tree on a platform is at rest

But when the same passenger is passing away in a train through station, observes that tree is in motion. In both conditions observer is right.

But observations are different because in first situation observer stands on a platform, which is reference frame at rest and in second situation observer moving in train, which is reference frame in motion.

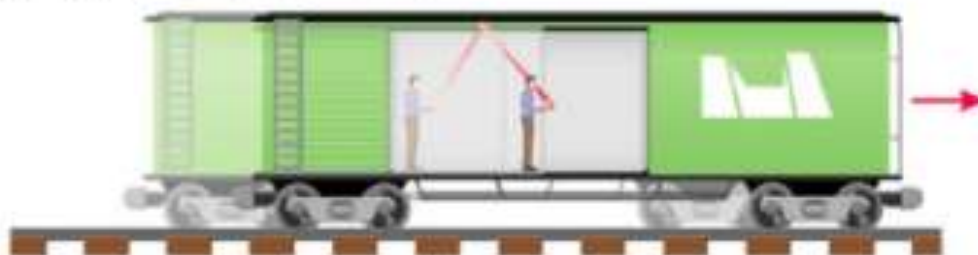
Inside the train, the ball goes up and down.



Outside the train, the ball appears to be going faster: It has the same up-and-down speed, plus the forward speed of the train.

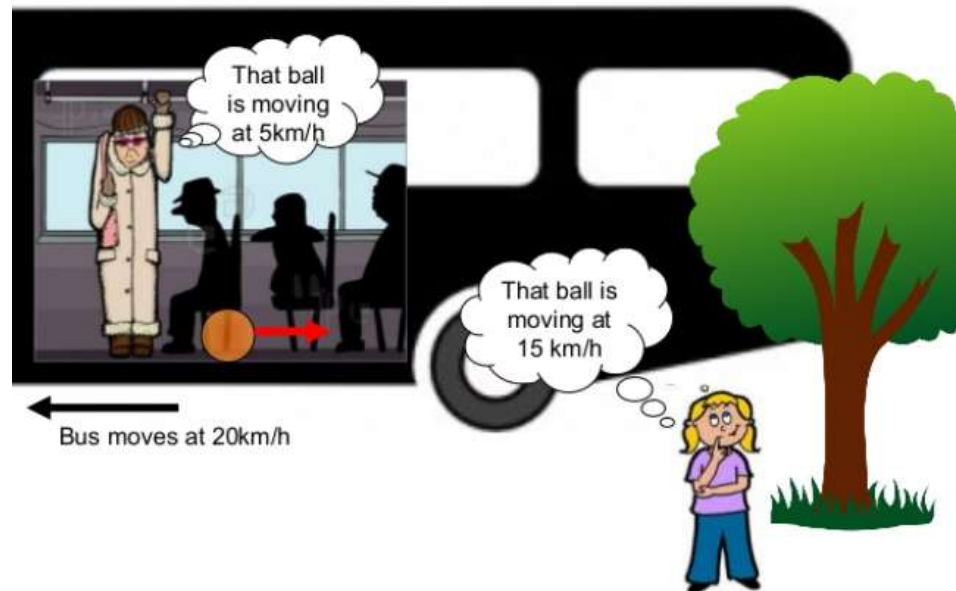


The faster the train is moving, the faster the ball appears to be going to the outside observer.



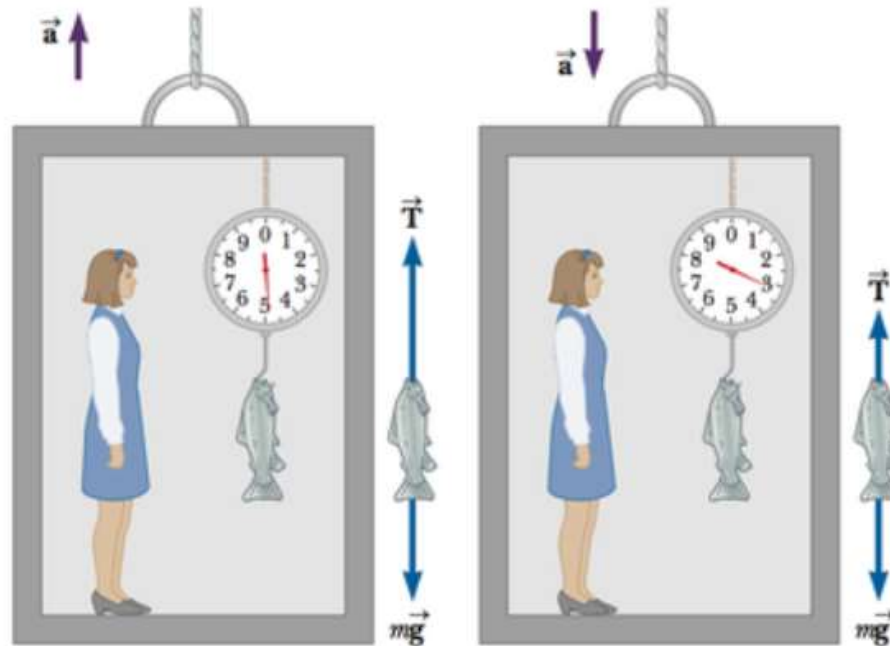
# Inertial and Non-Inertial Reference Frames

- Let's change things a little...
- Imagine that you are travelling on the bus with a ball beside you sitting on the floor. The bus is moving forward at a constant velocity of 20km/h.



When the elevator accelerates upward, the spring scale reads a value greater than the weight of the fish.

When the elevator accelerates downward, the spring scale reads a value less than the weight of the fish.



This figure will help you in the real world if you ever wanted to buy a fish and save some money. If a person weights a fish of mass  $m$  on a spring scale attached to the ceiling of an elevator, the scale will read different for when the elevator accelerates either upward or downward, but yet remain the same if the elevator moves with constant velocity.

**There are two types of frames of reference:-**

**1. Inertial frame of reference**

**2. Non – inertial frame of reference**

| S. No. | Inertial Frame of Reference                                                                                                                                                                                                                                                      | Non – inertial Frame of Reference                                                                                                                                                            |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.     | If in a given system a free object does not experience any acceleration or a force, the system is called the <b><i>inertial frame</i></b> . Any reference of frame that moves with the constant velocity relative to an inertial frame of reference is itself an inertial frame. | If we consider a free object in an accelerated frame, the object experiences acceleration or a force applied on it. Such a frame of reference is called <b><i>non – inertial frame</i></b> . |
| 2.     | Newton's laws are found to be valid in these frames of references.                                                                                                                                                                                                               | Newton's laws are not valid.                                                                                                                                                                 |
| 3.     | <b>Example:</b> - A passenger in a car, moving with uniform velocity experiences no force on him.                                                                                                                                                                                | <b>Example:</b> - A passenger in a car which is accelerating, experiences a push backward though no external force is applied on him.                                                        |

## Special Theory of Relativity:

The two basic *postulates* on which the theory rests may be stated in a precise form thus:

- (i) **The laws of physics all take the same identical form for all frames of reference in uniform relative motion, i.e., for all inertial frame of reference.**

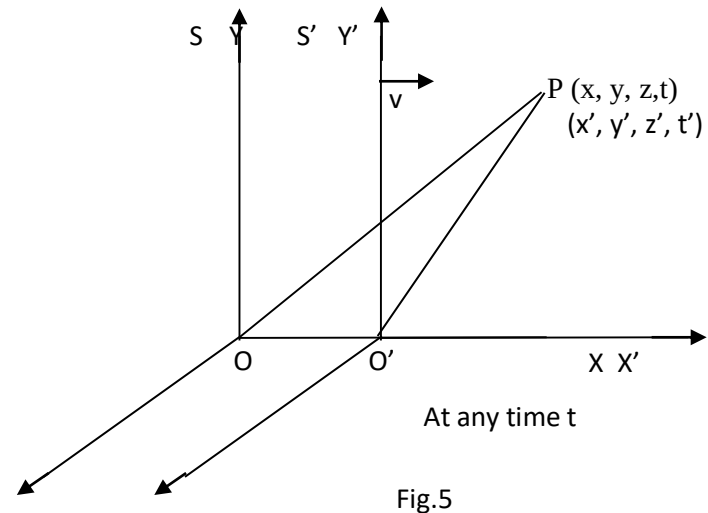
This postulate follows from the absence of universal frame of reference. If the laws of physics were different for different observers in relative motion, the observers could find from these differences which of them were “stationary” in space and which were “moving”. But such a distinction does not exist, and the principle of relativity expresses this fact.

- (ii) This postulate is, also called as *principle of equivalence*.

**The speed of light is constant in vacuum for all observers, i.e., speed of light is a universal constant.** The speed of light in vacuum neither depends on the direction of light nor the motion of the source and the observer. In other words, the speed of light in vacuum has the same value in all directions in all inertial frames of reference.

# Lorentz Transformations :

Consider two inertial frames of references  $S$  and  $S'$ . The frame  $S'$  is moving with uniform velocity  $v$  w.r.t.  $S$  along  $+X$  direction such that  $Y$  and  $Z$  axes of both frames remains parallel to each other as shown in fig.5



Let  $t$  and  $t'$  are the times of an event measured by the observer sitting at the origin  $O$  and  $O'$  of both frames respectively. For convenience let us assume that at  $t = t' = 0$  the origins  $O$  and  $O'$  of frames  $S$  &  $S'$  are coincident.

Let a pulse of light is generated at time  $t = 0$  at the origin which grows in the space. Now consider the situation when the pulse reaches at pt.  $P$ . Let  $(x, y, z, t)$  and  $(x', y', z', t')$  be the positions and time coordinates of  $P$  measured by observers  $O$  and  $O'$  in frames  $S$  and  $S'$  respectively. When the pulse is observed from  $S$ , then we have

$$\text{Velocity of light} = \text{distance} / \text{time}$$



Velocity of light = distance /time

$$c = \frac{OP}{t} = \frac{(x^2 + y^2 + z^2)^{1/2}}{t}$$

$$\Rightarrow x^2 + y^2 + z^2 - c^2 t^2 = 0 \quad \dots\dots\dots (1)$$

When the pulse is observed from S', then we have

$$c = \frac{OP'}{t'} = \frac{(x'^2 + y'^2 + z'^2)^{1/2}}{t'}$$

$$\Rightarrow x'^2 + y'^2 + z'^2 - c^2 t'^2 = 0 \quad \dots\dots\dots (2)$$

Moreover  $y = y'$  and  $z = z'$  (3)

From equations (1) & (2), using equation (3) we have

$$x^2 - c^2 t^2 = x'^2 - c^2 t'^2 \quad \dots\dots\dots (4)$$

The transformation between x and x' can be represented by the simple relationship

$$x' = K (x - vt) \quad \dots\dots\dots (5)$$

where K being independent of x and t.

If we suppose that the system S is moving relative to S' with velocity - v along +X direction (because motion is relative), then

$$x = K (x' + vt') \quad \dots\dots\dots (6)$$

Substituting the value of x' from equation (5) in equation 6 we have

$$x = K [K(x - vt) + vt']$$

$$\frac{x}{K} = K (x - vt) + vt' \quad \text{or} \quad vt' = \frac{x}{K} - K (x - vt)$$

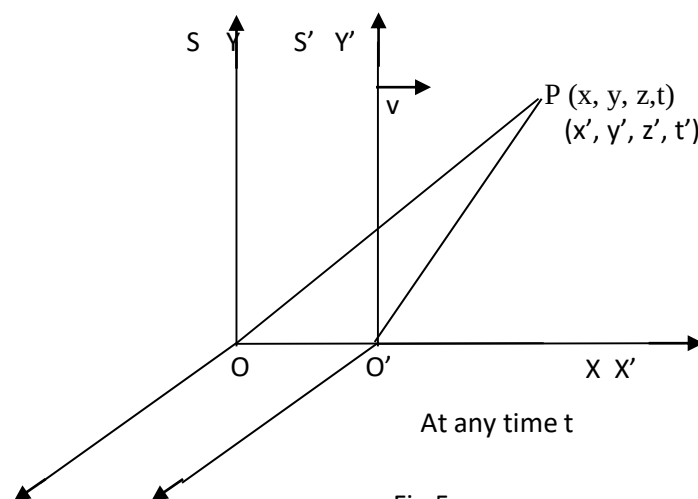


Fig.5

$$\underline{vt'} = \frac{x}{K} - K(x-vt)$$

or

$$t' = \frac{x}{Kv} - \frac{K(x-vt)}{v} = \frac{x}{Kv} - \frac{Kx}{v} + Kt$$

$$t' = K[t - \frac{x}{v} (1 - \frac{1}{K^2})] \dots\dots\dots (7)$$

Substituting the value of  $x'$  from equation (5) and  $t'$  from equation (7) in equation (4), we get

$$\underline{x^2 - c^2 t^2} = K^2 (x - vt)^2 - c^2 K^2 [t - \frac{x}{v} (1 - \frac{1}{K^2})]^2 \dots\dots\dots (4)$$

$$\underline{x^2 - c^2 t^2} - K^2 (x - vt)^2 + c^2 K^2 [t - \frac{x}{v} (1 - \frac{1}{K^2})]^2 = 0$$

$$\underline{x^2 - c^2 t^2} - K^2 (x^2 - 2xvt + v^2 t^2) + c^2 K^2 [t^2 + \frac{x^2}{v^2} (1 - \frac{1}{K^2})^2 - \frac{2xt}{v} (1 - \frac{1}{K^2})] = 0$$

\dots\dots\dots (8)

This is an identity and hence the coefficients of various powers of  $x$  and  $t$  must vanish separately.

$$\text{Equating the coefficients of } t^2 \text{ to zero, we get}$$

$$-c^2 - K^2 v^2 + c^2 K^2 = 0 \dots\dots\dots (9)$$

$$\text{or } K^2(c^2 - v^2) = c^2 \quad \text{or} \quad K^2 = c^2 / (c^2 - v^2)$$

$$\text{or } K^2 = \frac{1}{\{1 - (v^2/c^2)\}} \quad \text{or} \quad K = \frac{1}{\sqrt{\{1 - (v^2/c^2)\}}} \dots\dots\dots (10)$$

Substituting the value of  $K$  from equation (10) in equation (5), we have the Lorentz Transformation of space i.e.,

$$\underline{x'} = \frac{x - vt}{\sqrt{(1 - v^2/c^2)}} \dots\dots\dots (A)$$

Substituting the value of K from equation (10) in equation (5), we have the Lorentz Transformation of space i.e.,

$$x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}} \quad \dots\dots\dots(A)$$

Similarly, substituting the value of K in equation (7), we have the Lorentz Transformation for time i.e.

$$t' = K \left[ t - \frac{x}{v} \left( 1 - \frac{1}{K^2} \right) \right] \quad \dots\dots\dots (7)$$

$$t' = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \left[ t - \frac{x}{v} \left\{ 1 - \frac{(c^2 - v^2)}{c^2} \right\} \right]$$

or

$$t' = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \left[ t - \frac{xv^2}{vc^2} \right]$$

or

$$t' = \frac{t - \frac{xv}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \dots(B)$$

Equations (A) and (B) are called as Lorentz transformation equations.

$$y = y', \quad z = z'$$

# Inverse Lorentz transformation equation

If we assume that the system S is moving with velocity  $-v$  relative to  $S'$  along positive direction of X, Then the Lorentz transformation equations can be expressed as

$$x = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}}, \quad y = y', \quad z = z'$$
$$t = \frac{t' + \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

These are known as inverse Lorentz transformation equations.

# Special cases of L.T. equitation's

$$x' = \gamma (x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = \gamma \left( t - \frac{vx}{c^2} \right)$$

$$\text{Lorentz factor } \gamma = \left( \sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}$$

It can be easily seen that if  $v \ll c$  then  $(v/c) \rightarrow 0$ , then

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t$$

These are **Galilean transformations**. Thus Lorentz Transformations reduce to Galilean transformations if  $v \ll c$ .

Now let us take two events in frame S defined by  $(x_1, y_1, z_1, t_1)$  &  $(x_2, y_2, z_2, t_2)$ . These two events are called *simultaneous* if  $\Delta t = t_2 - t_1 = 0$ .

Now if  $\Delta x = (x_2 - x_1) = 0$ ,  $\Delta y = (y_2 - y_1) = 0$  &  $\Delta z = (z_2 - z_1) = 0$  i.e. the two events are occurring at the same position then they are called as *colocal*.

And if the two events occur at the same position ( $\Delta x = 0, \Delta y = 0, \Delta z = 0$ ) and at the same time ( $\Delta t = 0$ ) then they are called *coincident*.

Now Differentiating Lorentz equations, we have

$$x' = \gamma(x - vt)$$

$$y' = y$$

$$z' = z$$

$$t' = \gamma \left( t - \frac{vx}{c^2} \right)$$

$$\text{Lorentz factor } \gamma = \left( \sqrt{1 - \frac{v^2}{c^2}} \right)^{-1}$$

$$\Delta x' = \frac{\Delta x - v\Delta t}{(1 - v^2/c^2)^{1/2}}$$

$$\Delta y' = \Delta y$$

$$\Delta z' = \Delta z$$

$$\Delta t' = \frac{\Delta t - v\Delta x/c^2}{(1 - v^2/c^2)^{1/2}}$$

From these equations it follows that if  $\Delta x = 0$ ,  $\Delta y = 0$ ,  $\Delta z = 0$ , but  $\Delta t \neq 0$  then  $\Delta x'$  is not zero i.e. if two events are colocal in one inertial frame then it may not be colocal in another frame of reference (*Relativity of Colocality*).

Now again if  $\Delta t = 0$  but  $\Delta x \neq 0$ ,  $\Delta y \neq 0$ ,  $\Delta z \neq 0$  then  $\Delta t' \neq 0$ . Thus if two events are simultaneous in one inertial frame, they may not be simultaneous in another inertial frame (*Relativity of Simultaneity*).

If  $\Delta x = \Delta y = \Delta z = 0$  and also  $\Delta t = 0$ , then  $\Delta x' = \Delta y' = 0$ ,  $\Delta z' = 0$  and  $\Delta t' = 0$  i.e., if the two events are coincident in one inertial frame of reference, they will also be coincident in another inertial frame.

| Galilean Transformations                 | Lorentz Transformations<br>(Relativistic)                                                               | Inverse Lorentz Transformations<br>(Relativistic)                                                         |
|------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| $x' = x - vt$ $y' = y$ $z' = z$ $t' = t$ | $x' = \frac{x - vt}{\sqrt{1 - v^2/c^2}}$ $y' = y$ $z' = z$ $t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}}$ | $x = \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}$ $y = y'$ $z = z'$ $t = \frac{t' + vx'/c^2}{\sqrt{1 - v^2/c^2}}$ |

# Conclusions of Lorentz transformations

The Lorentz transformations are a set of equations that relate the space and time coordinates of events in different inertial frames of reference that move at a constant speed relative to each other. They are consistent with the postulates of special relativity, which state that the laws of physics are the same in all inertial frames and that the speed of light is constant in vacuum.

**Some of the conclusions of the Lorentz transformations are:**

- Length contraction:** The length of an object moving along its direction of motion is shorter than its length at rest, as measured by an observer in a different frame. The factor by which the length contracts is called the Lorentz factor, given by  $\gamma = \frac{1}{\sqrt{1 - v^2/c^2}}$ , where  $v$  is the relative speed of the frames and  $c$  is the speed of light.

- Time dilation:** The time interval between two events occurring at the same location in a moving frame is longer than the time interval measured by an observer in a stationary frame. The factor by which the time dilates is also given by the Lorentz factor.

- Relativity of simultaneity:** Two events that are simultaneous in one frame of reference may not be simultaneous in another frame. The difference in time between the events depends on the relative speed and distance between the frames.

- Velocity addition:** The velocity of an object measured in one frame of reference is not simply added to or subtracted from the relative velocity of another frame. Instead, a more complicated formula involving the Lorentz factor and the speed of light is used to find the velocity of the object in another frame.

- Mass-energy equivalence:** The mass of an object increases with its speed, as measured by an observer in a different frame. The increase in mass is related to the increase in kinetic energy by the famous equation  $E = mc^2$ , where  $E$  is the energy,  $m$  is the mass, and  $c$  is the speed of light.

These conclusions have profound implications for our understanding of space and time, and challenge our common sense notions based on classical mechanics.

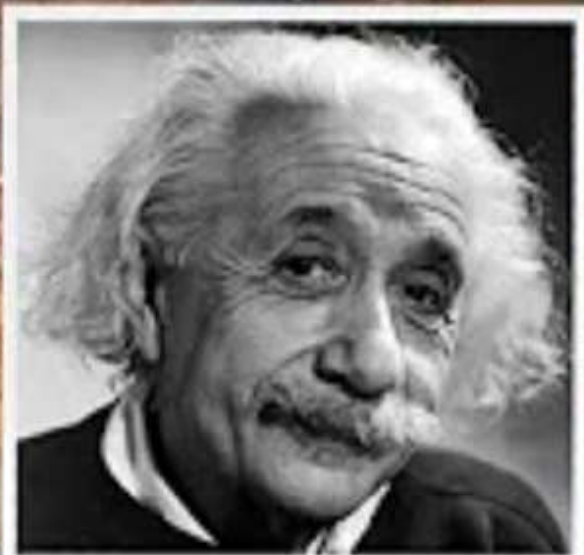


# Length Contraction

क्या है ?



$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$



*Albert Einstein*

## The phenomenon of Length contraction is Allegedly Real

The body of the ruler or any length will be shortened due to the transformation of the equations. The higher the velocity, the more shortening there will be. Relativists maintain that this effect is physical and real.

Proper length  $L_0$



Contracted length  $L$



## LENGTH CONTRACTION : RELATIVITY OF LENGTH

The length  $l$  of an object in motion with respect to an observer always appears to the observer to be shorter than its length  $l'$  when it is at rest with respect to him. **This contraction occurs only in the direction of the relative motion.** Length contraction can be derived as follows :

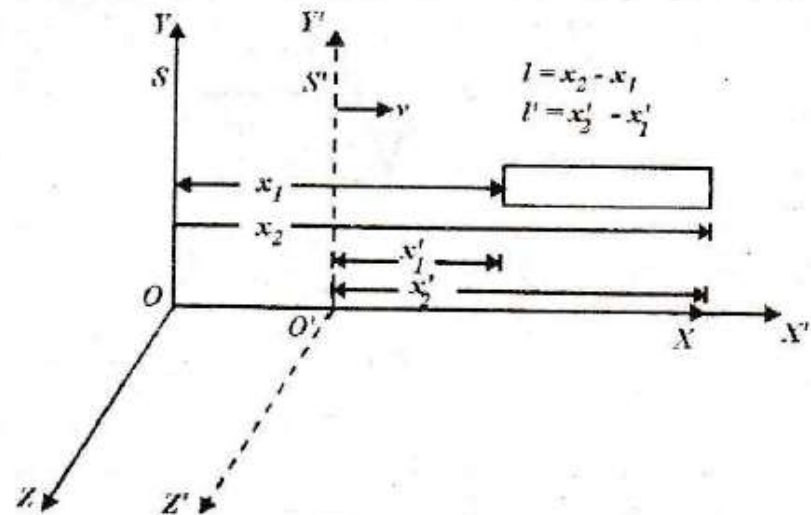


Fig.

Consider two coordinate systems S and S'. The S' moving with velocity  $v$  with respect to S along the +ve X-axis. For simplicity, let us suppose that the axis of two coordinate systems coincide at  $t = t' = 0$ .

Let a rod be placed in S' along X-axis. If  $x'_1$  and  $x'_2$  are coordinates of the abscissae of the ends of the rod, then its length  $l'$  in system S' is given by

$$l' = x'_2 - x'_1 \text{ (Proper length)} \quad \dots$$

Now let an observer in system S measures the length of the same rod. If  $x_1$  and  $x_2$  are the abscissae of the ends of the same rod at the same time  $t$ , then the length  $l$  of the rod is given by

$$l = x_2 - x_1 \text{ (Observed length)} \quad \dots$$

$$l' = x'_2 - x'_1 \text{ (Proper length)} \quad \dots (1)$$

$$l = x_2 - x_1 \text{ (Observed length)} \quad \dots (2)$$

From Lorentz transformation equations, we may write

$$x'_2 = \frac{x_2 - vt}{\sqrt{1 - v^2/c^2}}$$

and  $x'_1 = \frac{x_1 - vt}{\sqrt{1 - v^2/c^2}}$

Using these values of  $x'_1$  and  $x'_2$  in equation (1), we have

$$l' = x'_2 - x'_1 = \frac{x_2 - vt}{\sqrt{1 - v^2/c^2}} - \frac{x_1 - vt}{\sqrt{1 - v^2/c^2}}$$

$$l' = \frac{x_2 - x_1}{\sqrt{1 - v^2/c^2}}$$

$$l' = \frac{l}{\sqrt{1 - v^2/c^2}}$$

[Using equation (2)]

$$\Rightarrow l = l' \sqrt{\left(1 - \frac{v^2}{c^2}\right)}$$



$$\Rightarrow l = l' \sqrt{1 - \frac{v^2}{c^2}}$$

$$\text{Observed length} = \text{Proper length} \times \sqrt{1 - \frac{v^2}{c^2}}$$

Thus, the length of the rod, moving with velocity  $v$  with respect to the observer O,

is contracted by a factor  $\sqrt{1 - \frac{v^2}{c^2}}$  in the direction of motion.

This is known as **Length contraction** or **Lorentz-Fitzgerald contraction**.

The length  $l'$  measured in the frame in which rod is at rest is called "**proper length**".

The proper length ( $l'$  or  $L_0$ ) found in the rest frame is the maximum length which any observer will measure.

$$\Rightarrow l = l' \sqrt{1 - \frac{v^2}{c^2}}$$

$$\text{OR } L = L_0 \sqrt{1 - v^2/c^2}$$

$$\text{Observed length} = \text{Proper length} \times \sqrt{1 - \frac{v^2}{c^2}}$$

## Physical Significance:-

(i) when  $v \ll c$ , then  $v^2/c^2 \ll 1$ ,  $l = l'$ , OR ( $L = L_0$ )

i.e., the Observed length ( $L$  or  $l$ ) according to observer "O" is the same as that when it is at rest.

(ii) when  $v$  is comparable to  $c$ ,  $\sqrt{1 - v^2/c^2} < 1$  and in this case  $l < l'$  OR ( $L < L_0$ ) i.e., the apparent length  $L$  or  $l$  of a body in a frame  $S$  relative to which the rod is moving with uniform velocity  $v$  will be less than its proper length  $L_0$  or  $l'$  by the factor  $\sqrt{1 - v^2/c^2}$ . This contraction of length of a body along the motion is called **Lorentz – Fitzgerald contraction**.

If  $V=0.5C$ , So  $\sqrt{1 - v^2/c^2} = \sqrt{1 - [0.5c]^2/c^2} = \sqrt{1 - [0.5]^2} = \sqrt{0.75} = 0.866$

So  $L = L_0 (0.866) \Rightarrow$  if  $L_0 = 10\text{m}$ , so  $L = 8.66\text{m}$

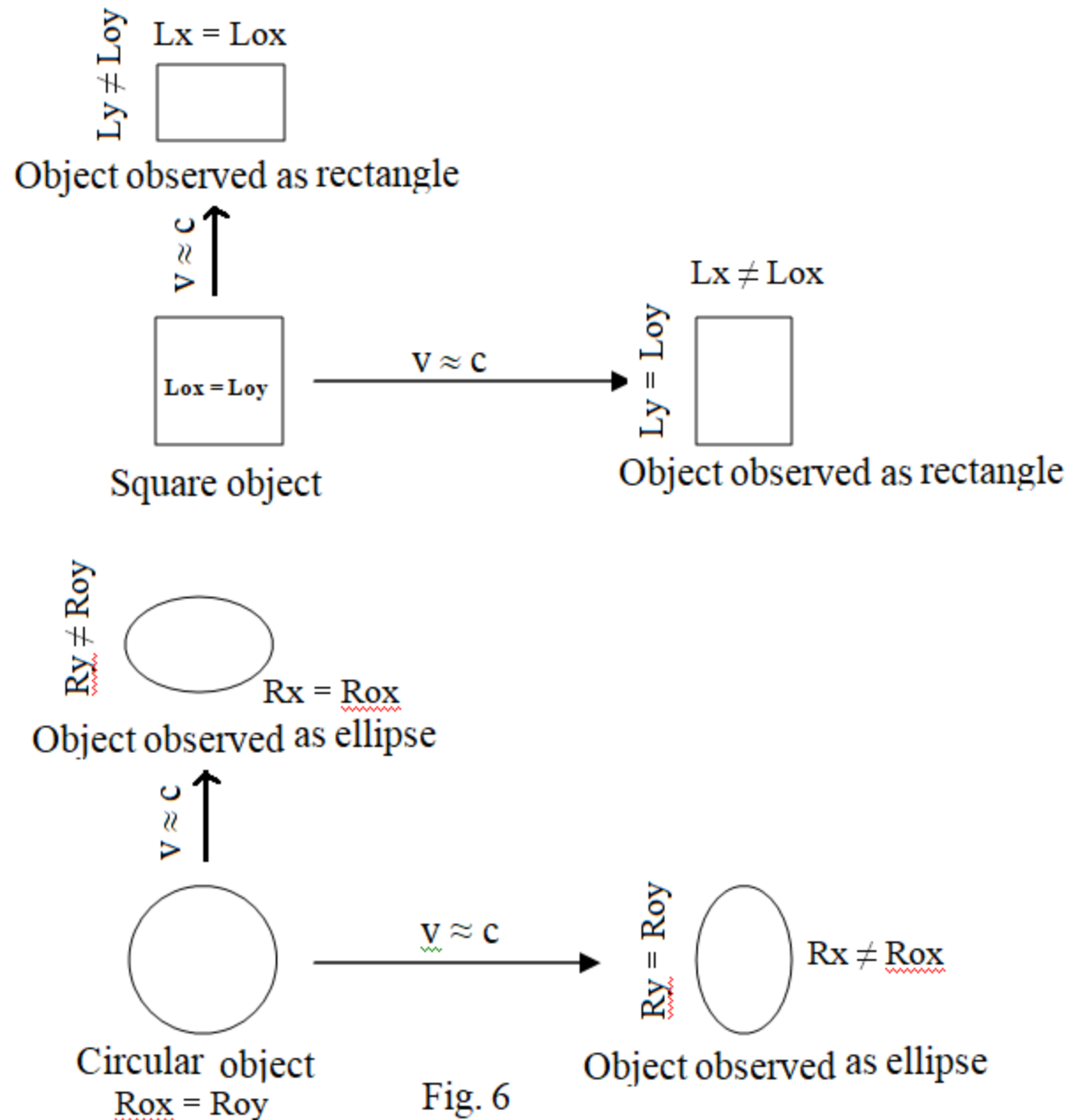
(iii) when  $v \geq c$ ,  $v^2/c^2 \geq 1$ , so that  $l = 0$ , OR ( $L = 0$ ) or imaginary i.e., That is not possible. **This implies that no material body can have a velocity equal to or greater than the velocity of the light.**

# Contraction of length of a body along the motion

If the length is measured in a direction normal to the direction of motion no contraction is observed as  $y' = y$  and  $z' = z$  in the Lorentz transformations.

Thus a moving square will appear as a rectangle with shorter side along the direction of motion and

a circle will appear as an ellipse with its minor axis along the direction of motion.



## Length Contraction Depends on the velocity of Object



$$V = 0$$



$$\xrightarrow{\quad} \\ V = 0.3C$$



$$\xrightarrow{\quad} \\ V = 0.6C$$



$$\xrightarrow{\quad} \\ V = 0.9C$$

Reference:-Video Lecture on Length contraction:-

<https://www.youtube.com/watch?v=p6hhFkSXpRY>

## NUMERICAL PROBLEMS ON LENGTH CONTRACTION

A rod has length 100 cm. When the rod is in a satellite moving with a velocity that is one half of the velocity of light relative to laboratory, what is the length of the rod as determined by an observer (a) in the satellite and (b) in the laboratory.

**Sol.** Given that proper length  $l' = 100$  cm,  $v = \frac{1}{2} c \Rightarrow \frac{v}{c} = 0.5$

(a) The observer in the satellite is at rest relative the rod hence the length of the rod as measured by him is 100 cm.

(b) The length  $l$  of the rod in the laboratory is given by

$$\begin{aligned} l &= l' \sqrt{1 - \frac{v^2}{c^2}} \\ &= 100 \sqrt{1 - (0.5)^2} = 100 \sqrt{0.75} \\ &= 85.6 \text{ cm} \end{aligned}$$



**Calculate the percentage contraction of a rod moving with a velocity of  $0.8c$  in a direction inclined at  $45^\circ$  to its own length.**

**Sol.** Given that  $v = 0.8c \Rightarrow \frac{v}{c} = 0.8$

Angle between velocity & length of rod  $= \theta = 45^\circ$

Let  $L_0$  be the length of the rod at rest, then length's components along and  $\perp$  to the direction of motion are  $L_0 \cos 45^\circ$  &  $L_0 \sin 45^\circ$  respectively.

$\therefore$  The apparent length along the direction of motion

$$L_{||} = (L_0 \cos 45^\circ) \sqrt{1 - (0.8)^2}$$

$$L_{||} = L_0 \times \frac{1}{\sqrt{2}} \times \sqrt{1 - 0.64}$$

$$L_{||} = L_0 \times \frac{1}{\sqrt{2}} \times 0.6 = \frac{0.6}{\sqrt{2}} L_0$$

and the apparent length  $\perp$  to the direction of motion

$$L_{\perp} = L_0 \sin 45^\circ = L_0 \left( \frac{1}{\sqrt{2}} \right)$$

$\therefore$  The length of the moving rod is  $L = \sqrt{L_{||}^2 + L_{\perp}^2}$

∴ The length of the moving rod is  $L = \sqrt{L_{||}^2 + L_{\perp}^2}$

$$L = \sqrt{\left[\left(\frac{0.6}{\sqrt{2}} L_0\right)^2 + (\sqrt{2} L_0)^2\right]} = \sqrt{\frac{1.36}{1.434}} = 0.825$$

∴ The percentage contraction in length of rod

$$= \frac{L_0 - L}{L_0} \times 100$$

$$= \frac{L_0 - 0.825 L_0}{L_0} \times 100 = 0.175 \times 100 = 17.5\%$$

$$= 17.5\%$$

**Calculate the length and the orientation of a meter rod in a frame of reference which is moving with a velocity equal to  $0.6c$ , in a direction making an angle  $30^\circ$  with the rod.**

**Sol.** Given that  $l = 1 \text{ m}$ ,  $v = 0.6c \Rightarrow \frac{v}{c} = 0.6$

The angle between direction of motion & rod  $\theta = 30^\circ$

$\therefore$  The length components along and  $\perp$  direction of motion are

$$l'_{||} = l \cos 30^\circ \text{ \& \; } l_{\perp} = l \sin 30^\circ$$

$\therefore$  The apparent length along the moving frame is

$$l'_{||} = l_{||} \sqrt{1 - \frac{v^2}{c^2}} = (l \cos 30^\circ) \sqrt{1 - (0.6)^2}$$

$$l'_{||} = 1 \times \frac{\sqrt{3}}{2} \times \sqrt{1 - 0.36} = \frac{\sqrt{3}}{2} \times 0.8 = 1.732 \times 0.4 \text{ m}$$

and the apparent length  $\perp$  to direction of motion

$$l'_{\perp} = l_{\perp} = l \sin 30^\circ = \frac{1}{2} \text{ m} = 0.5 \text{ m}$$

Let this rod is now observed to make an angle  $\theta'$  with the direction of motion, then

$$\tan \theta = \frac{l'_{\perp}}{l'_{\parallel}} = \frac{0.5}{1.732 \times 0.4} = 0.72$$

$$\theta = \tan^{-1} (0.72)$$

**A spaceship of length 100 m takes  $4 \mu\text{s}$  to pass an observer on earth. What is its velocity relative to earth ?**

**Sol.** Given that

$$l' = 100 \text{ m}, \quad t = 4 \mu\text{s} = 4 \times 10^{-6} \text{ sec}, \quad v = ?$$

Fly by time of a spaceship for an observer is

$$t = \frac{l}{v} = \frac{l'}{v} \sqrt{(1 - v^2/c^2)} = l' \left( \frac{1}{v^2} - \frac{1}{c^2} \right)^{1/2}$$

$$\frac{1}{v^2} = \frac{1}{c^2} + \left( \frac{t}{l'} \right)^2$$

$$\frac{1}{v^2} = \frac{1}{(3 \times 10^8)^2} + \left( \frac{4 \times 10^{-6}}{100} \right)^2 = 16.11 \times 10^{-16}$$

$$\Rightarrow v = 2.49 \times 10^7 \text{ m/s}$$







# Assignments



- **YOU ALL HAVE TO WATCH “INTERSTELLAR MOVIE” BEFORE ATTENDING MY NEXT LECTURE**
- OR**
- <https://www.youtube.com/watch?v=fSzDm19LVIg>



Interstellar (2014) - IMDb  
imdb.com



Interstellar movie review & film ...  
rogerebert.com



Interstellar (2014) | Movie...  
popmoviee.com



Interstellar (2014) - Rotten ...  
rottentomatoes.com



The Science of Interst...  
space.com



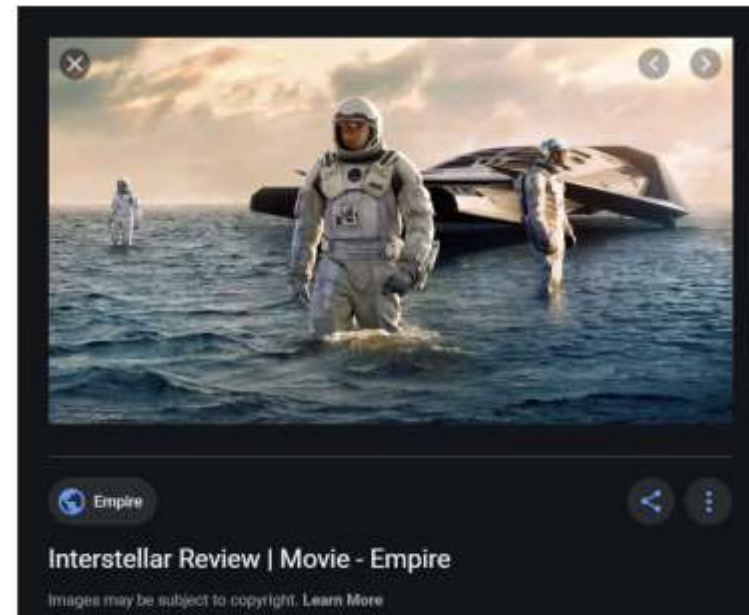
Interstellar (2014) - IMDb  
imdb.com



Interstellar Review | Movie - Empire  
empireonline.com



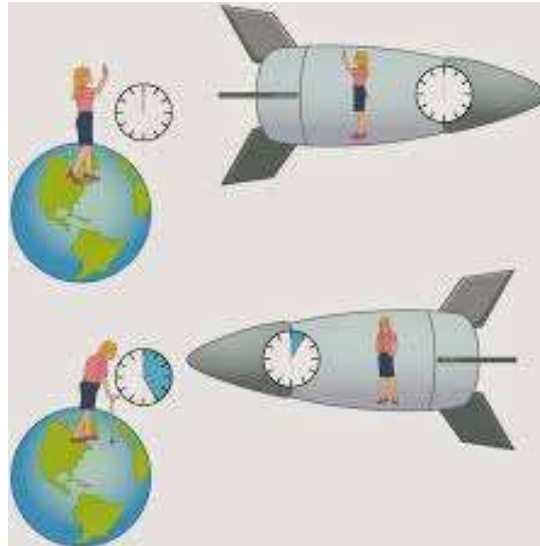
Interstellar\* on Behanc...  
pinterest.com





## (ii) Time dilation:-

**Time dilation** is the lengthening of the time interval between two events for an observer in an inertial frame that is moving with respect to the rest frame of the events (in which the events occur at the same location).



### **TIME DILATION : RELATIVITY OF TIME**

Time dilation means to **lengthen (or to become larger)** an interval of time. Measurements of time intervals are affected by relative motion between an observer and the object (what is observed). As a result, **a clock that moves with respect to an observer ticks more slowly or to have slowed down by a factor  $\left(\frac{1}{\sqrt{1 - v^2/c^2}}\right)$ , than it does without such motion.**

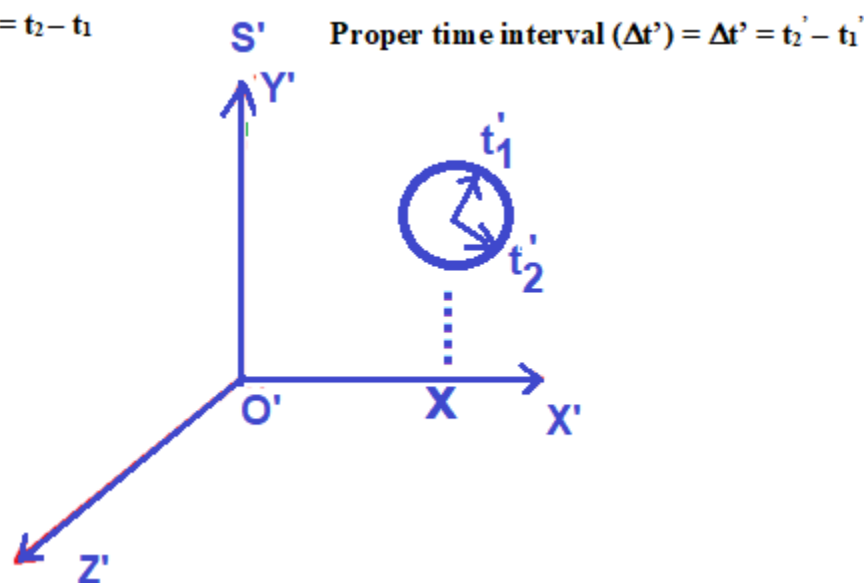
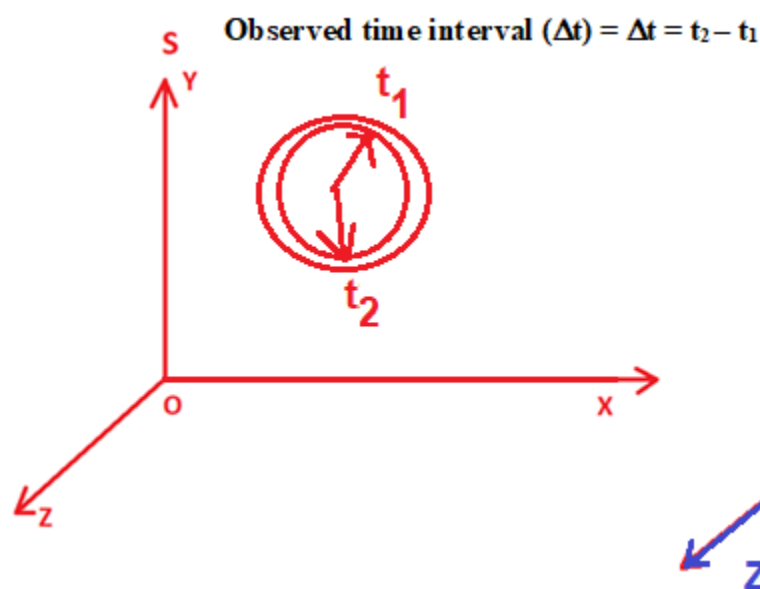
**Time Dilation can be derived as follows :**

Consider two systems  $S$  and  $S'$ .  $S'$  is moving with velocity  $v$  in the  $X$ -axis. For simplicity, the two reference frames coincide at  $\Delta t' = \Delta t = 0$ . Let a clock is placed at the point ' $x$ ' in moving frame  $S'$ . An observer  $O'$  in  $S'$  finds that the clock gives two ticks at times  $t_1'$  and  $t_2'$ . The time-interval between the ticks in  $S'$  in which the clock is at rest is  $\Delta t'$ , i.e.,

$$\text{Proper time interval } (\Delta t') = \Delta t' = t_2' - t_1' \quad \dots\dots (1)$$

Now, suppose another observer  $O$  measures the time interval (with his own clock) between the same two ticks from a stationary frame of reference  $S$ . If it records the same ticks at times  $t_1$  and  $t_2$ , the time interval appears in frame  $S$  is  $\Delta t$ , i.e.,

$$\text{Observed time interval } (\Delta t) = \Delta t = t_2 - t_1 \quad \dots (2)$$



According to inverse Lorentz transformation

$$t_1 = \frac{t_1' + vx_1'/c^2}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{Proper time interval } (\Delta t') = \Delta t' = t_2' - t_1' \quad \dots\dots (1)$$

$$\text{Observed time interval } (\Delta t) = \Delta t = t_2 - t_1 \quad \dots\dots (2)$$

and

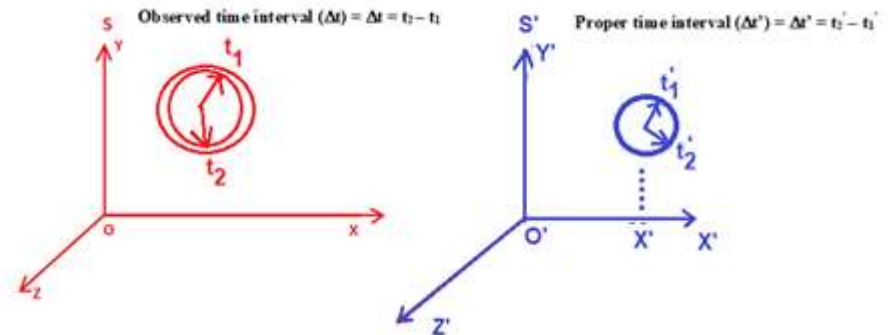
$$t_2 = \frac{t_2' + vx_2'/c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Using these values of  $t_1$  and  $t_2$  in equation (2), we get

$$\Delta t = t_2 - t_1 = \frac{\left(t_2' + \frac{x_2'v}{c^2}\right) - \left(t_1' + \frac{x_1'v}{c^2}\right)}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Delta t = \frac{t_2' - t_1'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}}$$



from Equation (1),  $\Delta t' = t_2' - t_1'$

It shows that the time interval  $\Delta t$  appears to be lengthened by a factor  $\frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$  in stationary observer in frame S. The proper time interval is the

time-interval measured by a clock attached to the moving body. In the above example  $\Delta t'$  is the proper time interval and  $\Delta t$  is observed time interval.

**Salient features**

$$\Delta t = \frac{\Delta t'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

If  $v=0.5c$ , So  $\sqrt{1 - v^2/c^2} = \sqrt{1 - [0.5c]^2/c^2}$   
 $= \sqrt{1 - [0.5]^2} = \sqrt{0.75} = 0.866$   
So  $\Delta t = \Delta t' / (0.866) \Rightarrow$  if  $\Delta t' = 10\text{Sec}$ , so  $\Delta t = 11.54\text{sec}$

- (i) When  $v$  is very small compared to  $c$ , the factor  $\frac{v^2}{c^2}$  will be negligibly small compared to unity, hence  $\Delta t = \Delta t'$  i.e., the time interval recorded by a moving clock is the same as that when it is at rest.
- (ii) When  $v$  is comparable to  $c$ ,  $\left( \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \right)$  will be more than unity, therefore  $\Delta t > \Delta t'$  i.e., the time interval between two events recorded by a moving clock appears to be greater than the time interval between the same events recorded by the clock when it is at rest.
- (iii) When  $v$  is equal to  $c$  or greater than  $c$ ,  $\frac{v^2}{c^2}$  will equal or greater than unity, therefore the time interval  $\Delta t$  becomes infinite or imaginary. This implies that no material body can have the velocity equal to or greater than the velocity of light.
- (iv) This is a real effect that applies not only to clocks based on light beams but to time itself; all clocks run more slowly according to an observer in relative motion, biological clocks included. Even the growth, aging, and decay of living systems are slowed by the time dilation effect.

The time-dilation effect predicted by special relativity has been accurately confirmed by observations of the increased lifetime of unstable elementary particles traveling at nearly the speed of light.

### Experimental verification of time dilation:

#### **(1) μ - meson theory:**

The μ - mesons are created by fast cosmic rays at a height of 10 km above the earth in large numbers. They have a speed of about 0.998c and the life time of a μ - meson is 2.2μs. So the distance covered by a μ - meson in this lifetime will be  $2.2 \times 10^{-6} \times 0.998 \times 3 \times 10^8 = 658.68$  meters before they disintegrate or decay into an electron and two neutrinos each. And yet they are found copiously in the laboratory 10 km below the spot where they are formed.

How is this possible? The answer is supplied by special theory of relativity. The μ - meson has a life time of  $t_0 = 2.2 \times 10^{-6}$  sec. in its own frame of reference on earth, however, the life time is elongated, due to relativistic effect, to the value t given by

$$t = \frac{t_0}{(1-v^2/c^2)^{1/2}} = \frac{2.2 \times 10^{-6}}{\sqrt{1 - (0.998)^2}}$$
$$= 34.38 \times 10^{-6} \text{ sec.}$$

which is more than 15 times that in their own frame of reference. In this life time a μ - meson can travel a distance

$$= 0.998c \times 34.38 \times 10^{-6} = 10.29 \text{ km}$$

This at once explains their presence in the laboratory, 10 km below, at the ground level.



## **(2) Decay of $\pi^+$ mesons (pions):**

A direct confirmation of time dilation is provided by a decay of  $\pi^+$  mesons (pions). These particles are unstable and disintegrate at an exponential rate in a proper frame in which they are at rest. Their half-life is  $1.8 \times 10^{-8}$  s (mean life is  $2.5 \times 10^{-8}$  s). Beams of  $\pi$  mesons with a velocity nearly  $0.99c$  can be produced by bombarding the target of an accelerator with high-energy protons. With a proper half life  $\tau_0 = 1.8 \times 10^{-8}$  s, 50% of the pions are expected to disappear within this time, if they were at rest. If we assume their velocity to be nearly equal to the speed of light  $c$ , they would traverse a distance  $3 \times 10^8 \times 1.8 \times 10^{-8} = 5.4$  m before their number decays to half the original number in their proper frame. In laboratory a flux of  $\pi$  mesons was measured at two positions 30m apart. Actual laboratory measurements show that nearly 40% of  $\pi$  mesons decayed after traveling a distance of 30 m, but according to its half-life, 98% of  $\pi$  mesons should decay after traveling the same distance. This inconsistency was solved by assuming the time dilation of its half life in laboratory frame.

In laboratory frame the pions moving with a velocity  $0.99 c$  will have half-life (according to theory of relativity).

$$\begin{aligned}\tau &= \frac{\tau_0}{\sqrt{1 - v^2 / c^2}} = 7\tau_0 \\ &= 7 \times 1.8 \times 10^{-8} \text{ s} \\ &= 12.6 \times 10^{-8} \text{ s}\end{aligned}$$

Thus in the laboratory frame the observed distance traveled by pions should be about  $12.6 \times 10^{-8} \times 3 \times 10^8 = 37.8$  m before they decay by 50% or nearly 30 m before they decay by 40%. This verifies experimentally the time dilation.

**TWIN PARADOX:-** The twin paradox is a thought experiment in special relativity involving identical twins, one of whom makes a journey into space in a high-speed rocket and returns home to find that the twin who remained on Earth has aged more.

Twin A and B are 20 years each. Twin A travels towards a star 30 light years away at a speed  $0.8c$ . He then returns home. Twin B remain at the home. What are the ages of A and B as observed by A and as observed by B ? Are the two observations are different ?

Sol. Given that  $d = 30$  light-year

$$= 30 \times 3 \times 10^8 \times 365 \times 24 \times 60 \times 60 \text{m}$$

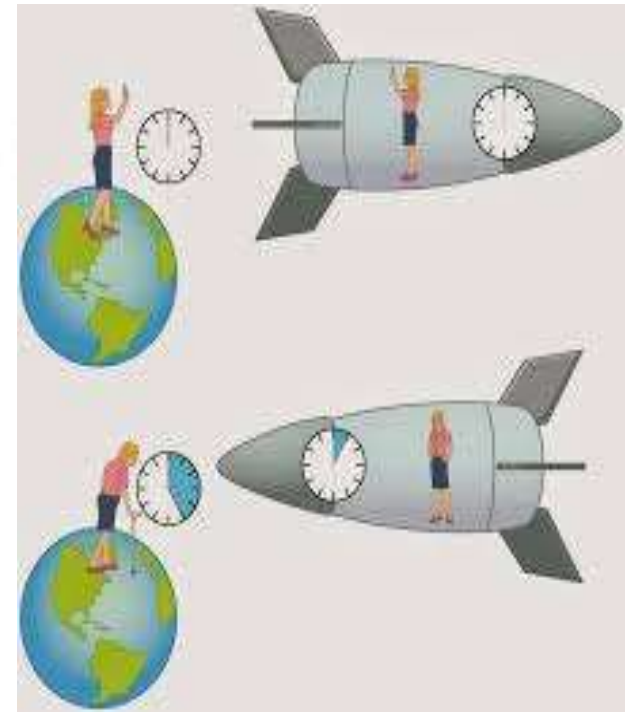
$$v = 0.80c \Rightarrow \frac{v}{c} = 0.8$$

The man B who stays at home, journey takes the time

$$t = \frac{2d}{v}$$

$$t = \frac{2 \times 30 \times 3 \times 10^8 \times 365 \times 24 \times 60 \times 60}{0.80 \times 3 \times 10^8} \text{ sec}$$

$$t = \frac{2 \times 30 \times 365 \times 24 \times 60 \times 60}{0.80 \times 365 \times 24 \times 60 \times 60} \text{ years}$$





The man B who stays at home, journey takes the time

$$t = \frac{2 \times 30}{0.8} = 75 \text{ Years}$$

The man A who makes the to and fro journey,  
the time interval appears to be

$$t' = t \sqrt{1 - \frac{v^2}{c^2}}$$

$$t' = 75 \text{ years} \times \sqrt{1 - (0.80)^2} = 45 \text{ Years}$$

Hence the Ages observed by A

Age of A = 20 + 45 = 65 Years

Age of B = 20 + 45 = 65 Year

and Ages observed by B

Age of A = 20 + 75 = 95 Years

Age of B = 20 + 75 = 95 Years

Note:- Ages observed by Third Person C

(The person knows the concept of relativity of time)

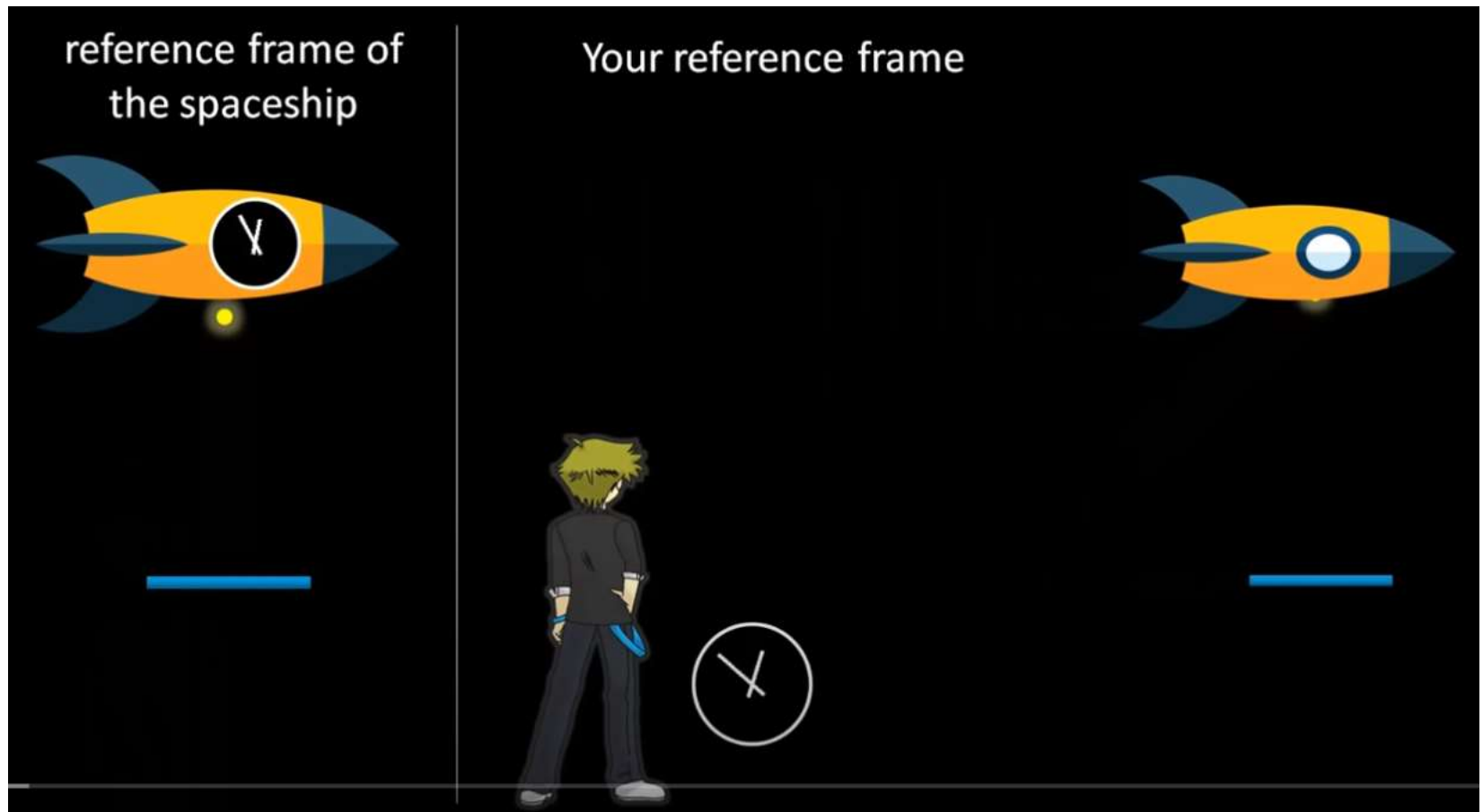
Age of A = 20 + 45 = 65 Years

Age of B = 20 + 75 = 95 Years

The twin paradox is a thought experiment in special relativity involving identical twins (A&B), one of whom makes a journey (A) into space in a high-speed rocket and returns home to find that the twin (B) who remained on Earth has aged more .i.e 95 Years old, but the A is 65Years old.

# Reference:-Video Lecture on Time dilation:-

<https://www.youtube.com/watch?v=d1sxag038IU>



**A spaceship of length 100 m takes  $4 \mu\text{s}$  to pass an observer on earth. What is its velocity relative to earth ?**

**Sol.** Given that

$$l' = 100 \text{ m}, \quad t = 4 \mu\text{s} = 4 \times 10^{-6} \text{ sec}, \quad v = ?$$

Fly by time of a spaceship for an observer is

$$t = \frac{l}{v} = \frac{l'}{v} \sqrt{1 - v^2/c^2} = l' \left( \frac{1}{v^2} - \frac{1}{c^2} \right)^{1/2}$$

$$\frac{1}{v^2} = \frac{1}{c^2} + \left( \frac{t}{l'} \right)^2$$

$$\frac{1}{v^2} = \frac{1}{(3 \times 10^8)^2} + \left( \frac{4 \times 10^{-6}}{100} \right)^2 = 16.11 \times 10^{-16}$$

$$\Rightarrow v = 2.49 \times 10^7 \text{ m/s}$$

A spacecraft is moving relative to earth. An observer on the earth finds that, according to her clock, 3601 sec. elapse between 1 P.M. and 2 P.M. on the spacecraft's clock. What is the space craft's speed relative to the earth ?

**Sol.** Given that proper time interval =  $t' = 3600$  sec (Total seconds between 1 P.M. to 2 P.M.) (on the earth)

Time interval (in moving frame with respect to the earth) =  $t = 3601$  sec.

$$\text{Time dilation relation } t = \frac{t'}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow \frac{v^2}{c^2} = 1 - \left(\frac{t'}{t}\right)^2$$

$$\Rightarrow v = c \sqrt{1 - \left(\frac{t'}{t}\right)^2}$$

$$\Rightarrow v = 2.998 \times 10^8 \text{ m/s} \sqrt{1 - \left(\frac{3600}{3601}\right)^2}$$

$$v = 7.07 \times 10^6 \text{ m/s}$$

A laboratory observation shows that a length of 2 metres is crossed by a beam of radioactive particle in time  $1.0 \times 10^{-8}$  sec. and in this process half the particle disintegrate. Calculate the proper half life of the particle. Also calculate the traversed length as seen by an observer fixed relative to particle.

Sol. Given that proper length  $= l' = 2 \text{ m}$

Improper half life  $= t = 1.0 \times 10^{-8} \text{ sec.}$

proper half life  $= t' = ?$  traversed length  $L = ?$

$\therefore$  The proper half life is given by time dilation relation

$$t' = t \sqrt{1 - \frac{v^2}{c^2}}$$

The event of disintegration occurs in the particles, whose speed in the laboratory is

$$v = \frac{l'}{t} = \frac{2 \text{ m}}{1.0 \times 10^{-8} \text{ sec}} = 2 \times 10^8 \text{ m/sec}$$

∴ The proper half life is given by time dilation relation

$$t' = t \sqrt{1 - \frac{v^2}{c^2}}$$
$$= (1.0 \times 10^{-8} \text{ sec.}) \sqrt{1 - \left(\frac{2 \times 10^8}{3 \times 10^8}\right)^2}$$

$$\Rightarrow t' = 1.0 \times 10^{-8} \text{ sec.} \times (0.78)$$

$$\Rightarrow t' = 0.78 \times 10^{-8} \text{ sec}$$

**traversed length  $L = ?$**

The path measured by the moving observer is

$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

$$L = 2 \text{ m} (0.78)$$

$$L = 1.56 \text{ m}$$





# Relativity of Velocities (Einstein's Velocity Addition Theorem):-

The Lorentz transformations enable us to transform the velocity from one frame of reference to another frame of reference moving with a relative constant velocity with respect to it.

Let  $u$  be the velocity of a particle in a proper frame of reference  $S$ . The components of this velocity are

$$u_x = \frac{dx}{dt}, u_y = \frac{dy}{dt}, u_z = \frac{dz}{dt} \quad \dots\dots\dots(1)$$

Another frame of reference  $S'$  which is initially coincident with frame  $S$ , moves with a uniform velocity  $v$  along  $+x$ -direction relative to frame  $S$ . At any instant  $t'$ , the coordinates of a point  $(x, y, z)$  are given by Lorentz transformations as

$$\begin{aligned} x' &= \gamma(x - vt) \\ y' &= y \\ z' &= z \\ t' &= \gamma(t - vx/c^2) \end{aligned} \quad \dots\dots\dots(2)$$

where  $\gamma = (1 - v^2/c^2)^{-1/2}$

Let  $u'$  be, the velocity of the same particle in this frame  $S'$  and its components are

$$u'_x = \frac{dx'}{dt'}, u'_y = \frac{dy'}{dt'}, u'_z = \frac{dz'}{dt'} \quad \dots\dots\dots(3)$$

Taking differentials of transformation equations (2), we have

$$dx' = \gamma(dx - vdt) = \gamma(u_x - v) dt \quad \dots\dots\dots(4)$$

$$dy' = dy \quad \dots\dots\dots(5)$$

$$dz' = dz \quad \dots\dots\dots(6)$$

$$\begin{aligned} dt' &= \gamma\left(dt - \frac{vdx}{c^2}\right) \\ &= \gamma\left(1 - \frac{vu_x}{c^2}\right) dt \end{aligned} \quad \dots\dots\dots(7)$$

Taking differentials of transformation equations (2), we have

$$dx' = \gamma (dx - v dt) = \gamma (u_x - v) dt \quad \dots\dots\dots (4)$$

$$dy' = dy \quad \dots\dots\dots (5)$$

$$dz' = dz \quad \dots\dots\dots (6)$$

$$dt' = \gamma \left( dt - \frac{v dx}{c^2} \right) \\ = \gamma \left( 1 - \frac{v u_x}{c^2} \right) dt \quad \dots\dots\dots (7)$$

In equations (4) and (7), we put  $dx = u_x dt$ . Dividing equations (4), (5) and (6) by equation (7) one gets

$$u_x' = \frac{dx'}{dt'} = \frac{(u_x - v)}{\left( 1 - \frac{v u_x}{c^2} \right)} \quad \dots\dots\dots (8)$$

$$u_y' = \frac{dy'}{dt'} = \frac{u_y}{\gamma (1 - v u_x / c^2)} \quad \dots\dots\dots (9)$$

$$u_z' = \frac{dz'}{dt'} = \frac{u_z}{\gamma (1 - v u_x / c^2)} \quad \dots\dots\dots (10)$$

where  $\gamma = (1 - v^2 / c^2)^{-1/2}$

## Einstein's Velocity Addition Theorem

$$u_x' = \frac{dx'}{dt'} = \frac{(u_x - v)}{\left(1 - \frac{vu_x}{c^2}\right)}, \quad u_y' = \frac{dy'}{dt'} = \frac{u_y}{\gamma(1 - vu_x/c^2)}, \quad u_z' = \frac{dz'}{dt'} = \frac{u_z}{\gamma(1 - vu_x/c^2)}, \quad \gamma = (1 - v^2/c^2)^{-1/2}$$

Above set of equations is called Lorentz transformation of velocities or law of relativity of velocities.

### SPECIAL CASES:-

- (i) If  $u_x'$  be very much less than  $c$ , so that  $u_x'/c \rightarrow 0$ , the above expressions all reduce to those obtained by Galilean transformation, i.e.,  $u_x = u_x' + v$ ,  $u_y = u_y'$  and  $u_z = u_z'$ .
- (ii) For higher values of  $u_x'$ , the denominator in the expression for  $u_x$  becomes greater than 1 and hence the resultant velocity  $u_x$  is smaller than  $(u_x' + v)$ , the value given by Galilean transformation.
- (iii) If  $u_x' = c$ , i.e., if the velocity of the particle along the  $x'$  – axis in frame  $S'$  be equal to that of light (i.e., if the particle be a photon), we have

$$u_x = \frac{(u_x' + v)}{\left(1 + \frac{vu_x'}{c^2}\right)} \Rightarrow u_x = \frac{c + v}{1 + vc/c^2} = \frac{c + v}{1 + v/c} = c \quad \text{i.e., the particle has same velocity } c \text{ in frame } S.$$

Thus, this particular velocity  $c$  alone is invariant in the two frames of reference irrespective to their relative velocity  $v$ .

- (iv) If  $v = c = u_x'$ , then we have

$$u_x = \frac{c + c}{1 + c.c/c^2} = c$$

This shows that the addition of velocity of light to the velocity of light merely reproduces the velocity of light.

- (v) If a particle is moving with a velocity  $\underline{u}_x'$  in a frame  $S'$  moving with a velocity  $v$  with respect to frame  $S$ , then the components of velocity in frame  $S$  will be

$$u_x = \frac{dx}{dt} = \frac{(u_x' + v)}{\left(1 + \frac{vu_x'}{c^2}\right)} \quad u_y = \frac{dy}{dt} = \frac{u_y'}{\gamma(1 + vu_x'/c^2)} \quad u_z = \frac{dz}{dt} = \frac{u_z'}{\gamma(1 + vu_x'/c^2)}$$

where  $\gamma = (1 - v^2/c^2)^{-1/2}$

- (vi) Einstein velocity addition theorem in conformity with the principle of constancy of speed of light: Consider a light photon moving along +x direction in the inertial frame  $S$  with velocity  $c$ , then  $\underline{u}_x = c$ ;  $\underline{u}_y = 0$  and  $\underline{u}_z = 0$

From Einstein's velocity addition theorem,

$$u_x' = \frac{dx'}{dt'} = \frac{(u_x - v)}{\left(1 - \frac{vu_x}{c^2}\right)} \quad u_y' = \frac{dy'}{dt'} = \frac{u_y}{\gamma(1 - vu_x/c^2)} \quad u_z' = \frac{dz'}{dt'} = \frac{u_z}{\gamma(1 - vu_x/c^2)}$$

where  $\gamma = (1 - v^2/c^2)^{-1/2}$

$$\therefore u_x' = \frac{c - v}{1 - v/c} = c, \quad \underline{u}_y' = 0, \quad \underline{u}_z' = 0$$

$\therefore$  The velocity of photon in the inertial frame  $S' = c$ .

If a light photon is moving along +y direction with a velocity  $c$  in the inertial frame  $S$ , then

$$\underline{u}_x = 0, \underline{u}_y = c, \underline{u}_z = 0$$

The corresponding velocity components as observed in the inertial frame  $S'$  are

$$u_x' = \frac{u_x - v}{1 - u_x v/c^2} = -v, \quad u_y' = \frac{c}{\gamma(1 - u_x v/c^2)} = c(1 - v^2/c^2)^{1/2}, \quad \underline{u}_z' = 0$$

$\therefore$  The speed of light as measured in frame  $S'$

$$\underline{u}' = \sqrt{(u_x')^2 + (u_y')^2 + (u_z')^2} = \sqrt{v^2 + c^2 - v^2} = c$$

Similarly, if a light photon is moving along +z direction in frame  $S$  with a velocity  $c$ , then the speed of light measured in frame  $S'$  will also be equal to  $c$ .

**Explain that Einstein's velocity addition theorem in conformity with the Principle of constancy of the speed of light.**

**OR**

**Prove that the speed of light is independent of the inertial frame of reference and is, therefore a universal constant**

**A radioactive nucleus while moving with a velocity of  $0.2c$  in the lab-frame emits a  $\beta$ -particle. The  $\beta$ -particle moves with a speed  $0.6c$  relative to the nucleus. If it is emitted perpendicular to the direction of motion of the nucleus, what is the velocity of the  $\beta$ -particles in lab-frame ?**

**Solution :** Let us consider the nucleus moving along X-axis and frame  $S'$  is attached to the nucleus. The nucleus emits  $\beta$ -particle with a velocity  $0.6c$  along Y-axis in the frame  $S'$ .

$$\text{Given: } u_x' = 0; u_y' = 0.6c; \quad v = 0.2c,$$

Velocity component  $u_x$  of  $\beta$ -particle

$$u_x = \frac{u_x' + v}{1 + vu_x' / c^2}$$

$$u_x = \frac{0 + 0.2c}{1 + 0} = 0.2c$$

Velocity component  $u_y$  of  $\beta$ -particle

$$u_y = \frac{u_y' \sqrt{1 - v^2 / c^2}}{1 + vu_x' / c^2}$$

$$u_y = \frac{0.6c \sqrt{1 - (0.2)^2}}{1 + 0}$$

$$= 0.59c$$

Therefore, the velocity of  $\beta$ -particle

$$\vec{u} = \hat{i} u_x + \hat{j} u_y$$

$$= \hat{i} (0.2c) + \hat{j} (0.59c) \quad \text{Ans.}$$



**A radioactive atom moving at a speed  $0.3c$  in X-direction emits an electron having speed  $0.8c$  in rest frame of atom. Find speed of the electron if it is emitted (a) in X-direction, and (b) Y-direction.**

**Solution :** Let the frame of reference fixed on atom be indicated by  $S'$  and the lab. frame by  $S$ .

(a)  $\beta$ -particle is emitted in X-direction.

Given:  $u'_x = 0.8c$ ,  $u'_y = 0$ ,  
 $u'_z = 0$ ,  $v = 0.3c$

$$u_x = \frac{u'_x + v}{1 + u'_x v / c^2}$$

$$= \frac{0.8c + 0.3c}{1 + 0.8 \times 0.3}$$

$$= \frac{1.1c}{1.24} = 0.887c$$

$$u_y = \frac{u'_y \sqrt{1 - v^2 / c^2}}{1 + u'_x v / c^2} = 0$$

$$u_z = 0$$

Hence in the lab. frame  $\beta$ -particle will move in the X-direction with a velocity  $0.887c$ .

**Ans.**

(b)  $\beta$ -particle emitted in Y-direction.

$$u'_x = 0; \quad u'_y = 0.8c;$$

$$u'_z = 0; \quad v = 0.3c$$

$$u_x = \frac{u'_x + v}{1 + u'_x v / c^2} = \frac{0 + 0.3c}{1 + 0} = 0.3c$$

$$u_y = \frac{u'_y \sqrt{1 - v^2 / c^2}}{1 + u'_x v / c^2}$$

$$= \frac{0.8c \sqrt{1 - 0.09}}{1 + 0}$$

$$= 0.8c \sqrt{0.91} = 0.763c$$

$$u_z = 0$$

Resultant velocity

$$u = \sqrt{u_x^2 + u_y^2 + u_z^2}$$

$$= \sqrt{0.09c^2 + (0.763)^2 c^2}$$

$$= 0.82c$$

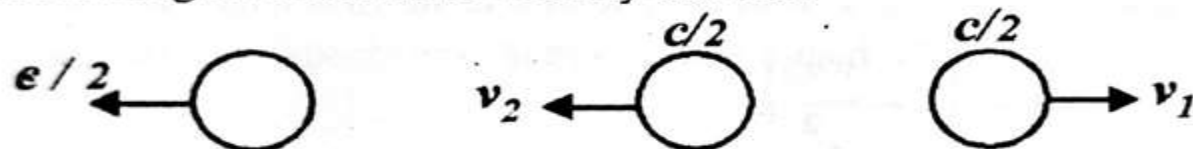
The direction of  $u$  in the x-y plane will be

$$\alpha = \tan^{-1} \frac{u_y}{u_x} = \tan^{-1} \frac{0.763c}{0.3c}$$

$$= \tan^{-1} (25.43) = 68.53^\circ$$

A huge missile explodes and divides into two equal parts, which take off with velocities  $\frac{c}{2}$ , one to the right and the other to the left. The piece that moves to the right again gets divided into two, by an explosion such that with respect to its own rest frame the resulting two pieces take off with velocities  $\frac{c}{2}$  to the right and left. Calculate the velocities of the last two pieces with respect to the earth.

Sol. Let  $v_1$  and  $v_2$  denote the velocities of the last two pieces with respect to the earth. According to relativistic velocity addition



$$u_x = \frac{u_x' + v}{1 + \frac{v u_x'}{c^2}}$$

For the first piece,

$$u_x = v_1, \quad u_x' = \frac{c}{2}, \quad v = \frac{c}{2}$$

$$\text{then } v_1 = \frac{\frac{c}{2} + \frac{c}{2}}{1 + \frac{\frac{c}{2} \times \frac{c}{2}}{c^2}} = \frac{c}{1 + 0.25} = \frac{4}{5}c$$

For the second piece

$$u_x = v_2, \quad u_x' = -\frac{c}{2}, \quad v = \frac{c}{2}$$

$$\text{then } v_2 = \frac{\frac{c}{2} - \frac{c}{2}}{1 + \frac{\left(\frac{c}{2}\right) \left(-\frac{c}{2}\right)}{c^2}} = 0$$

## Relativity of Mass :-

OR

## Relativistic Variation of Mass with Velocity:-

OR

## The Relation between Relativistic Mass(m) with Velocity (v) :-

### MASS INCREASES

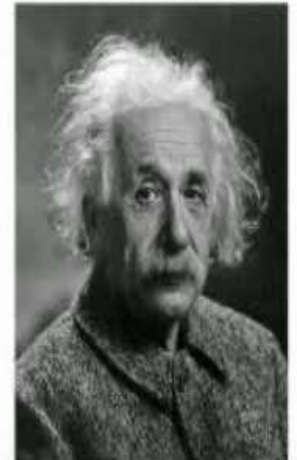
⇒ THE MASS OF AN OBJECT MOVING AT A SPEED  $v$  RELATIVE TO THE OBSERVER IS LARGER THAN ITS MASS WHEN AT REST RELATIVE TO THE OBSERVER

⇒ THE RELATIVISTIC MASS IS GIVEN AS

$$m = \frac{m'}{\sqrt{1 - v^2/c^2}}$$

where

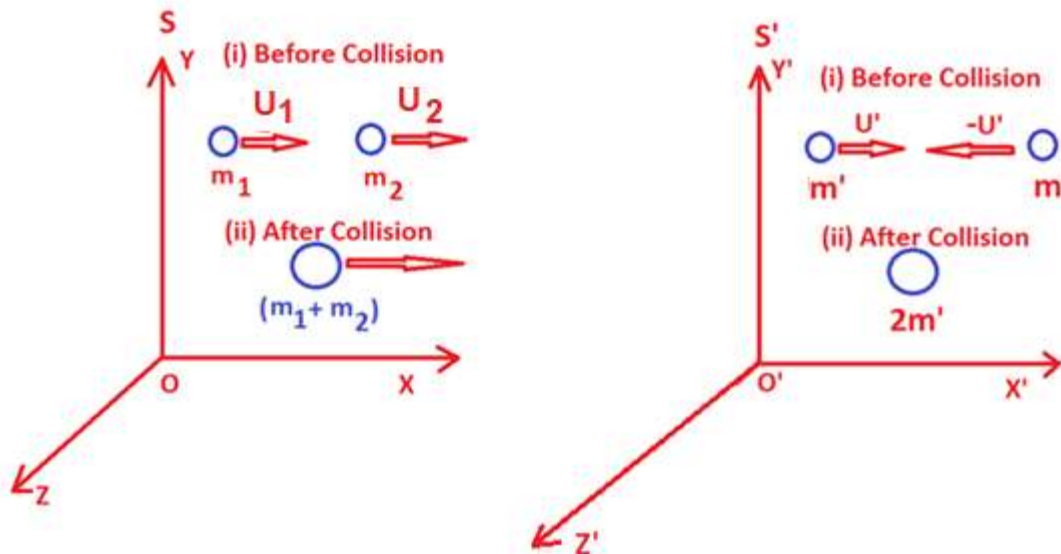
|      |                            |
|------|----------------------------|
| $m$  | relativistic mass          |
| $m'$ | mass of the object at rest |
| $v$  | speed of the moving frame  |
| $c$  | speed of light             |



## Relativity of Mass or Relativistic Variation of Mass with Velocity:

According to Newtonian mechanics the mass of a moving body is constant and independent of velocity. But the theory of relativity leads to a very different conclusion, viz, the variation of mass with velocity. To determine the velocity dependence of mass, consider a collision experiment of two identical, perfectly elastic smooth balls.

Suppose a frame  $S'$  is moving along  $+x$ -direction with a uniform velocity  $v$  relative to the frame  $S$ . In this inertial frame  $S'$  two bodies of equal masses  $m'$  traveling with velocities  $u'$  and  $-u'$  parallel to  $x$ -axis collide and after collision they coalesce into one body.



Hence by the **principle of conservation of momentum in frame  $S'$** .  
**momentum before collision = momentum after collision**

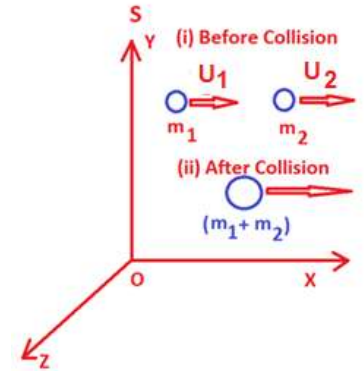
$$m'u' - m'u' = 2m'v', \Rightarrow 0 = 2m'v', \Rightarrow 0 = m'v', \Rightarrow v' = 0, \text{ Since } m' \neq 0$$

**Thus the velocity of coalesced body in frame  $S'$  will be zero i.e., Coalesced body is at rest in frame  $S'$ .**

Let us now consider how this collision experiment appears to an observer in frame S. The velocity of the two bodies will not appear to be equal for an observer in frame S as it is moving with a velocity  $-v$  relative to the frame S'. If the velocities of the two bodies are observed as  $u_1$  and  $u_2$  before collision in frame S, then by relativistic law of addition of velocities.

$$u_1 = \frac{u' + v}{1 + u'v/c^2} \quad \dots(1)$$

$$u_2 = \frac{-u' + v}{1 - u'v/c^2} \quad \dots(2)$$



If we assume that mass of bodies is variable, then the mass of the bodies in frame S will appear to be different as their velocities are different. Let  $m_1$  be the mass of the body traveling with a velocity  $u_1$  and  $m_2$  be the mass of the body traveling with a velocity  $u_2$ . After collision as the coalesced body is at rest with respect to frame S', it will appear to be moving with a velocity  $v$  in frame S and its mass will be  $(m_1 + m_2)$ .



If the velocities of the two bodies are observed as  $u_1$  and  $u_2$  before collision in frame S, then by relativistic law of addition of velocities.

$$u_1 = \frac{u'+v}{1+u'v/c^2} \qquad \dots(1)$$

$$u_2 = \frac{-u'+v}{1-u'v/c^2} \qquad \dots(2)$$

**According to first postulate of special theory of relativity, all physical laws are of the same form in all inertial frames, so the principle will also have the same form of conservation of linear momentum in frame S.**

**Momentum before collision = Momentum after collision,**

$$m_1u_1 + m_2u_2 = (m_1 + m_2)v$$

Substituting equations (1) and (2),

$$m_1\left(\frac{u'+v}{1+u'v/c^2}\right) + m_2\left(\frac{-u'+v}{1-u'v/c^2}\right) = (m_1 + m_2)v$$

or

$$m_1\left[\frac{u'+v}{1+u'v/c^2} - v\right] = m_2\left[v - \frac{-u'+v}{1-u'v/c^2}\right]$$

or

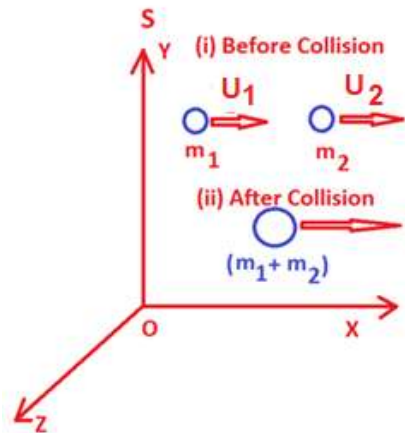
$$m_1\left[\frac{u'+v - v - u'v^2/c^2}{1+u'v/c^2}\right] = m_2\left[\frac{v - u'v^2/c^2 + u'-v}{1-u'v/c^2}\right]$$

or

$$\frac{m_1u'(1-v^2/c^2)}{1+u'v/c^2} = \frac{m_2u'(1-v^2/c^2)}{1-u'v/c^2}$$

∴

$$\frac{m_1}{m_2} = \frac{1+u'v/c^2}{1-u'v/c^2} \qquad \dots\dots\dots (3)$$





But from equation (1), we know that  $u_1 = \frac{u'+v}{1+u'v/c^2}$

$$\begin{aligned}
 1 - \frac{u_1^2}{c^2} &= 1 - \left[ \frac{u'+v}{1+u'v/c^2} \right] \frac{1}{c^2} \\
 1 - \frac{u_1^2}{c^2} &= 1 - \frac{u'^2/c^2 + v^2/c^2 + 2u'v/c^2}{1 + u'^2 v^2 / c^4 + 2u'v/c^2} \\
 1 - \frac{u_1^2}{c^2} &= \frac{1 + 2u'v/c^2 + u'^2 v^2 / c^4 - u'^2 / c^2 - v^2 / c^2 - 2u'v/c^2}{1 + u'^2 v^2 / c^4 + 2u'v/c^2} \\
 1 - \frac{u_1^2}{c^2} &= \frac{1 + u'^2 v^2 / c^4 - u'^2 / c^2 - v^2 / c^2}{(1 + u'v/c^2)^2} \\
 1 - \frac{u_1^2}{c^2} &= \frac{(1 - v^2 / c^2)(1 - u'^2 / c^2)}{(1 + u'v/c^2)^2} \dots(4)
 \end{aligned}$$

Similarly from equation (2), we know that  $u_2 = \frac{-u'+v}{1-u'v/c^2}$

$$1 - \frac{u_2^2}{c^2} = \frac{(1 - v^2 / c^2)(1 - u'^2 / c^2)}{(1 - u'v/c^2)^2} \dots(5)$$

$$\frac{m_1}{m_2} = \frac{1 + u'v/c^2}{1 - u'v/c^2} \dots (3) \quad 1 - \frac{u_1^2}{c^2} = \frac{(1 - v^2/c^2)(1 - u'^2/c^2)}{(1 + u'v/c^2)^2} \dots (4) \quad 1 - \frac{u_2^2}{c^2} = \frac{(1 - v^2/c^2)(1 - u'^2/c^2)}{(1 - u'v/c^2)^2} \dots (5)$$

So from equations (4) and (5), we have

$$\frac{(1 - u_2^2/c^2)}{(1 - u_1^2/c^2)} = \frac{(1 + u'v/c^2)^2}{(1 - u'v/c^2)^2}$$

$$\frac{\sqrt{(1 - u_2^2/c^2)}}{\sqrt{(1 - u_1^2/c^2)}} = \frac{(1 + u'v/c^2)}{(1 - u'v/c^2)} \dots (6)$$

From equations (3) & (6),

$$\frac{m_1}{m_2} = \frac{1 + u'v/c^2}{1 - u'v/c^2} \dots (3)$$

$$\frac{m_1}{m_2} = \frac{\sqrt{(1 - u_2^2/c^2)}}{\sqrt{(1 - u_1^2/c^2)}}$$

$$\frac{\sqrt{(1 - u_2^2/c^2)}}{\sqrt{(1 - u_1^2/c^2)}} = \frac{(1 + u'v/c^2)}{(1 - u'v/c^2)} \dots (6)$$

$$m_1 \sqrt{(1 - u_1^2/c^2)} = m_2 \sqrt{(1 - u_2^2/c^2)}$$

As both sides of this equation are independent to each other, it will be possible only when both sides are equal to some constant (say  $m_0$ ). In case of  $u = 0$ , this constant will have some finite value. This value is called rest mass of the particle.

$$\therefore m_1 \sqrt{(1 - u_1^2/c^2)} = m_2 \sqrt{(1 - u_2^2/c^2)} = m_0$$

$$\therefore m_1 = \frac{m_0}{\sqrt{(1 - u_1^2/c^2)}} \quad \text{and} \quad m_2 = \frac{m_0}{\sqrt{(1 - u_2^2/c^2)}}$$

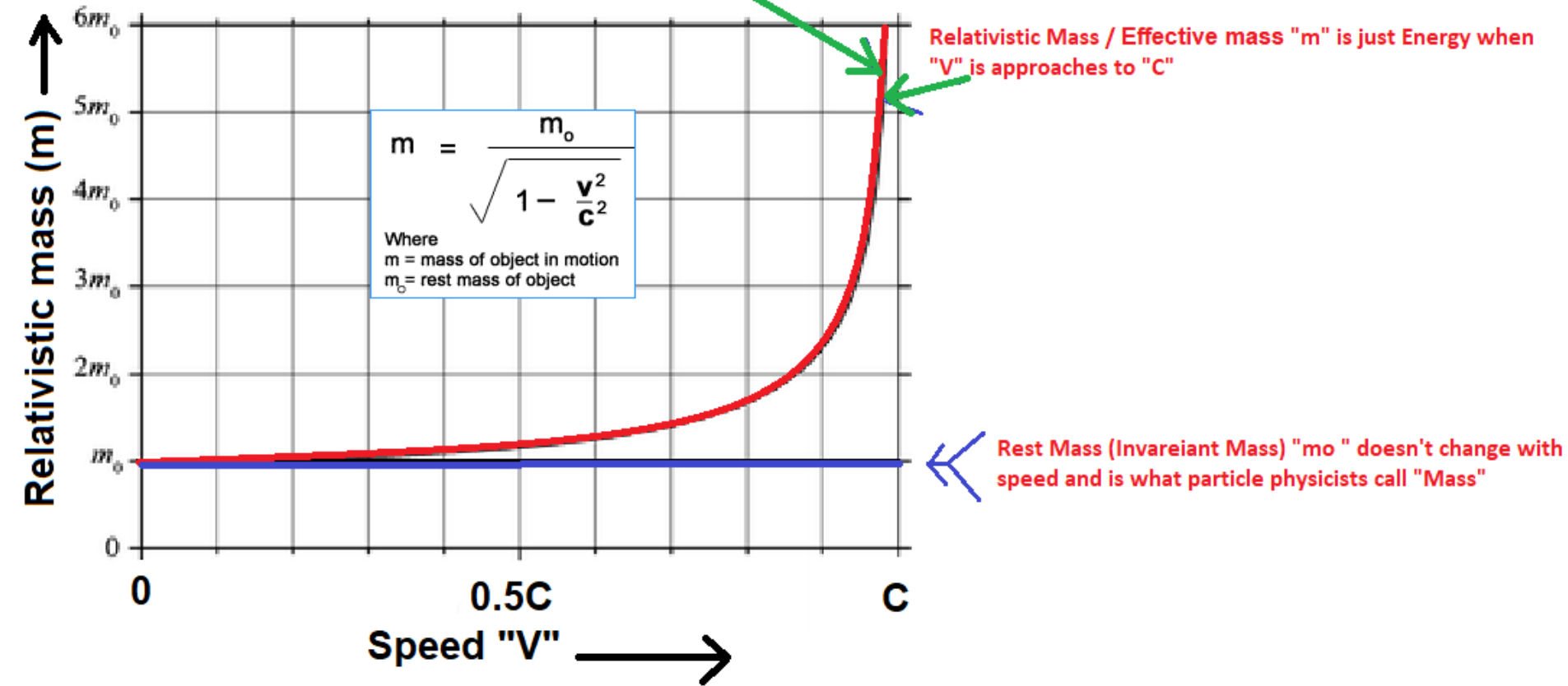
In general

$$m = \frac{m_0}{\sqrt{(1 - v^2/c^2)}} \dots (7)$$

The above equation shows the relation between effective mass ( $m$ ) of the particle of rest mass ( $m_0$ ) moving with a velocity  $v$ .

## The dependence of Relativistic mass $m$ of the particle on $V$ is shown in the following figure:-

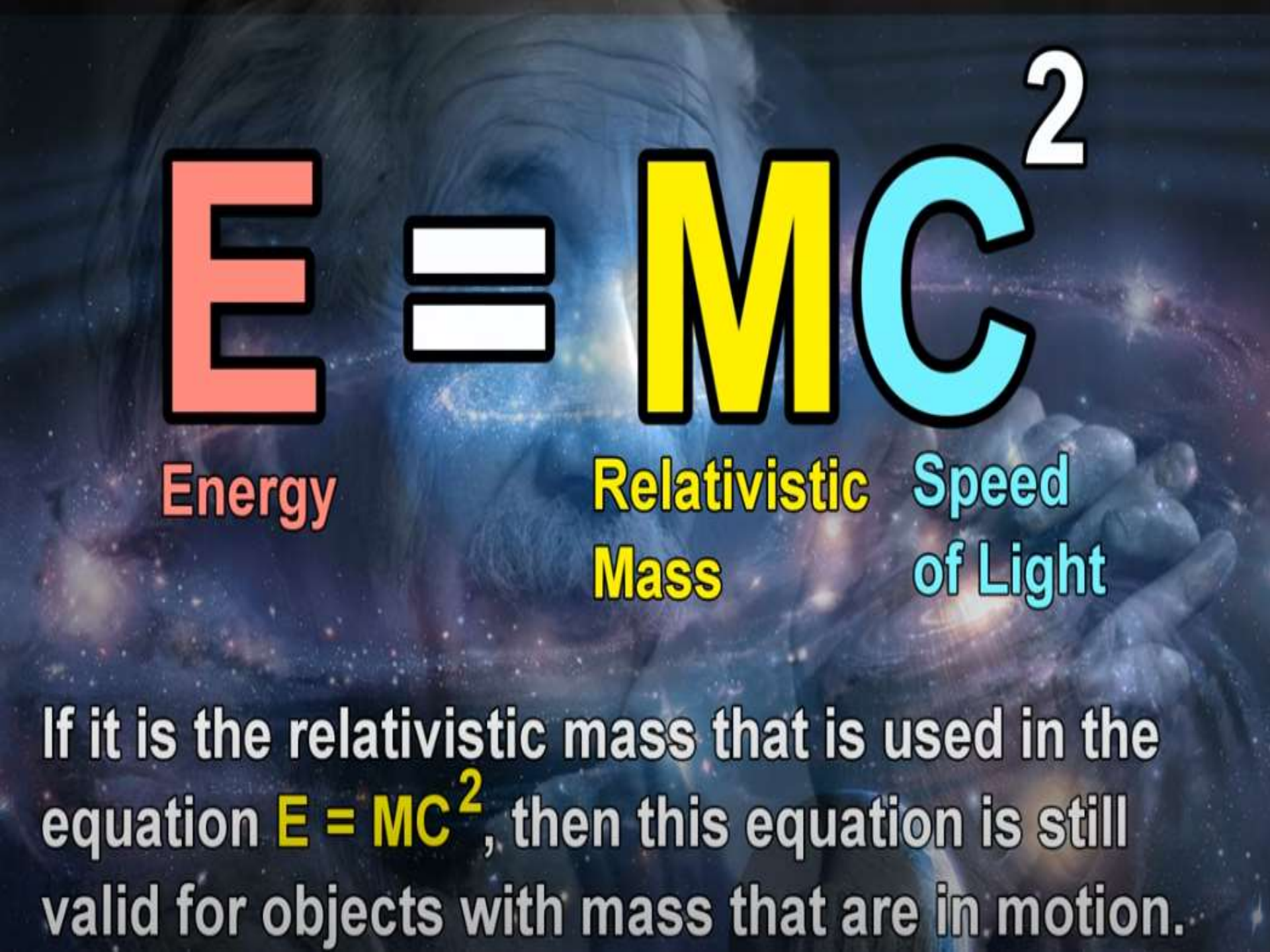
Mass is increasing very rapidly becoming infinite as " $V$ " is approaches to " $C$ "



### Physical Significance of Relativistic mass equation:-

It is evident from Relativistic mass or effective mass ( $m$ ) [equation (7)], that for small values of  $V$  i.e.,  $V \ll C$ ,  $V^2/C^2$  is negligible compared to 1 so that  $m \approx m_0$ . But as the velocity of the particle increases, the effective mass of the particle increases in comparison of its rest mass ( $m_0$ ).

Also **as  $V$  approaches to the speed of light  $C$ , the observed mass approaches infinity.**


$$E = MC^2$$

Energy

Relativistic  
Mass

Speed  
of Light

If it is the relativistic mass that is used in the equation  $E = MC^2$ , then this equation is still valid for objects with mass that are in motion.

# Mass-Energy Relation ( $E = mc^2$ ) :-

OR

## The Relation between Relativistic mass (m) and Energy (E):-

OR

## Relativistic Energy (E)-Momentum (P) Relation:-

In 1905 Einstein proposed a historic relation between mass and energy. According to this relation mass and energy can be considered as different manifestations of the same physical quantity. Thus a certain quantity of energy of any form (such as kinetic. potential. chemical. nuclear etc.) can be treated as mass or conversely.

**“Under appropriate circumstances mass can be converted into energy or vice-versa”.** Thus mass and energy are equivalent to each other.



According to Einstein's first postulate of relativity, all laws related with physical events have the same form in all inertial frames. Thus **Newton's second law and work'-energy theorem will remain valid in all inertial frames.**

**From work-energy theorem,**

**Increase in kinetic energy = work done by the particle**

$$\underline{dE_k} = \underline{dW} = \underline{Fdx}$$

From Newton's second law,

$$F = \frac{d(mv)}{dt}$$

$$\begin{aligned} dE_k &= \frac{d(mv)}{dt} dx = d(mv) \frac{dx}{dt} \\ &= v d(mv) \end{aligned}$$

$$\underline{dE_k} = \underline{mv dv} + v^2 dm \quad \dots(1)$$

Mass of the particle moving with a velocity  $v$ ,

$$m = \frac{m_0}{\sqrt{1 - v^2 / c^2}}$$

where  $m_0$  is the rest mass of the particle.

$$m^2 = \frac{m_0^2}{(1 - v^2 / c^2)}$$

or  $m^2 c^2 = m_0^2 c^2 + m^2 v^2$   $\Rightarrow m^2 = m_0^2 C^2 / (C^2 - V^2), \Rightarrow m^2 (C^2 - V^2) = m_0^2 C^2$

Differentiating it,

$$2mc^2 dm = 0 + 2vdvm^2 + 2mdmv^2$$

or  $c^2 dm = \underline{mv dv} + v^2 dm$  ....(2)

Comparing equations (1) & (2)

$$\underline{dE_k} = c^2 dm \text{ ....(3)}$$

When the force accelerates the body from rest to the velocity  $v$ , its mass increases from  $m_0$  to  $m$  (let) and its kinetic energy increases from 0 to  $\underline{E_k}$ .

Integrating equation (3).

$$\int_0^{E_k} dE_k = \int_{m_0}^m c^2 dm$$

$$\int_0^{E_k} dE_k = \int_{m_0}^m c^2 dm$$

$$\therefore \quad E_k = c^2(m - m_0) \quad \dots(4)$$

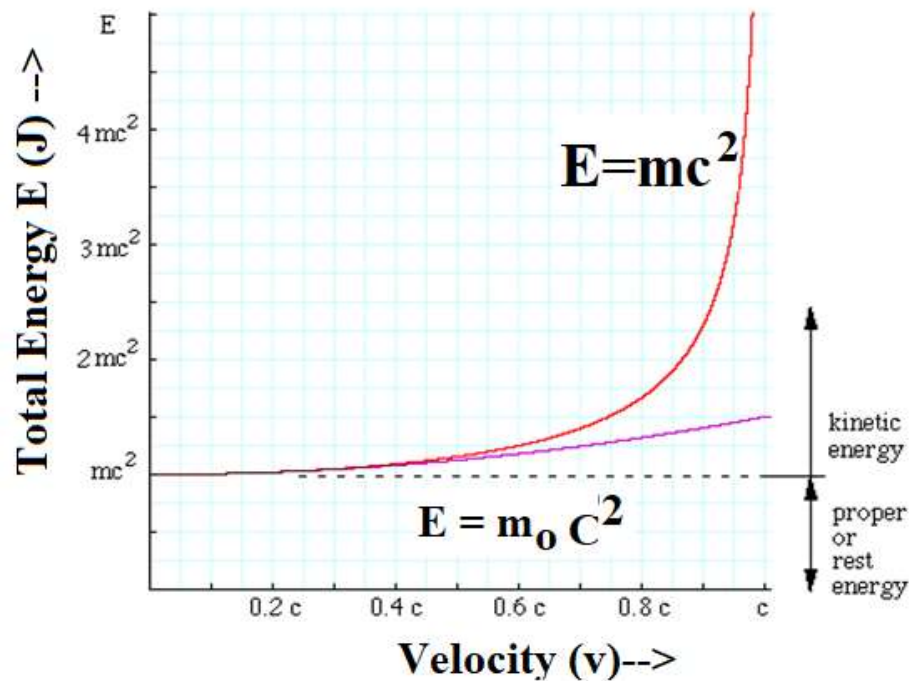
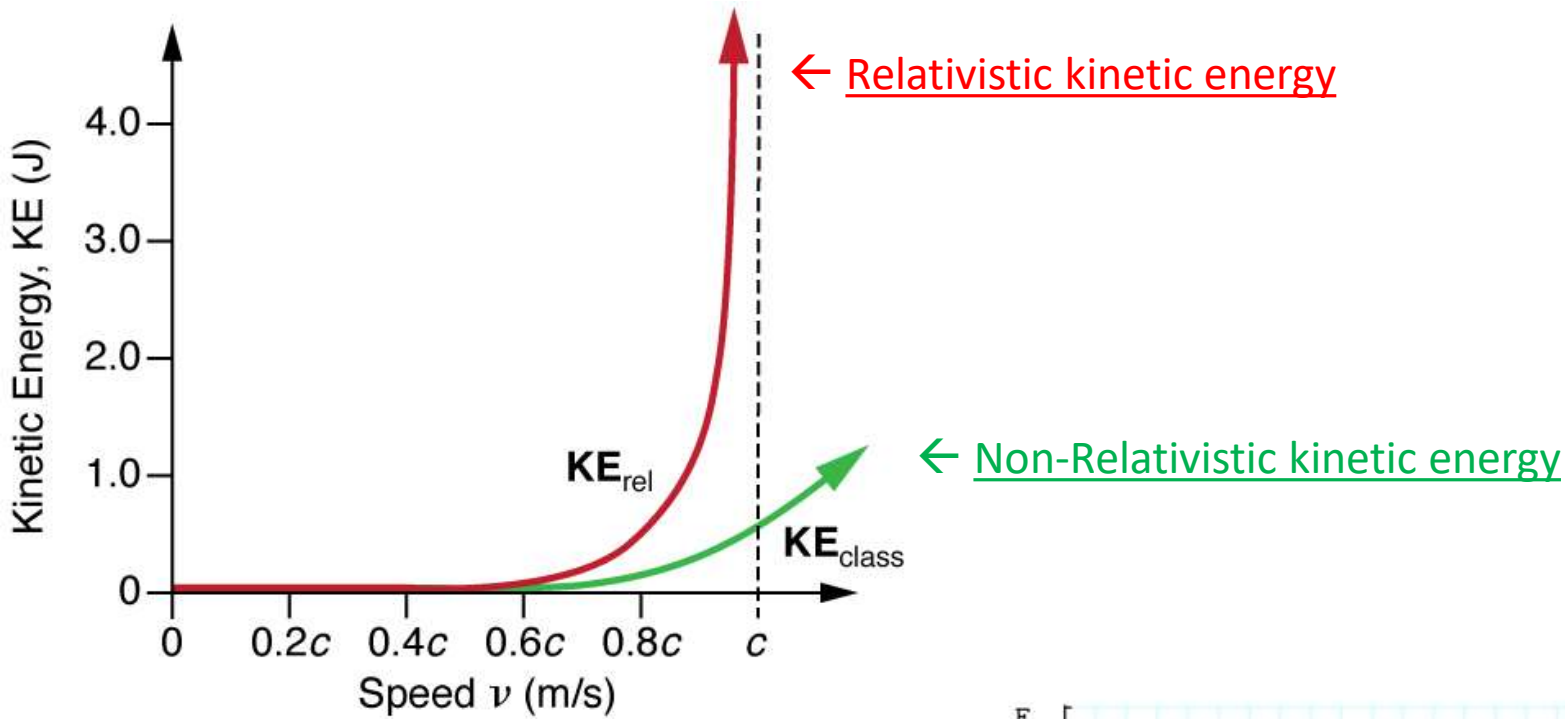
Since total energy of the particle is equal to the sum of its rest energy ( $m_0c^2$ ) and kinetic energy  $E_k$ . Thus

$$\begin{aligned} E &= E_k + m_0c^2 \\ &= c^2(m - m_0) + m_0c^2 \end{aligned}$$

$$E = mc^2 \quad \dots(5)$$

Where

E is the total energy of the particle, m is the relativistic mass of the particle, and c is the velocity of light, **this equation is called Einstein's mass-energy relation.**



(i) Non-relativistic kinetic energy of particle:

If the velocity of the particle is small compared to the velocity of light i.e.  $v \ll c$ , then from equation (4),

$$E_k = c^2(m - m_0) \quad \dots(4)$$

$$\begin{aligned} E_k &= c^2 m - c^2 m_0 = \frac{c^2 m_0}{\sqrt{1 - v^2 / c^2}} - c^2 m_0 \\ &= m_0 c^2 \left(1 - v^2 / c^2\right)^{-1/2} - m_0 c^2 \\ &= m_0 c^2 \left(1 + \frac{v^2}{2c^2} + \frac{3v^4}{8c^4} + \dots\right)^{-1/2} - m_0 c^2 \\ &= \frac{1}{2} m_0 v^2 + \frac{3}{8} m_0 v^2 \left(\frac{v^2}{c^2}\right) + \dots \end{aligned}$$

$$E_k = \frac{1}{2} m_0 v^2$$

Thus for small values of velocity  $v$ , equation (4) converts into classical formula of kinetic energy ( $E_k \approx m_0 v^2/2$ ).

(ii) Relativistic energy-momentum relation:

Total energy of a moving particle

$$E = mc^2 = \frac{m_0 c^2}{\sqrt{1 - v^2 / c^2}} \quad \dots(1)$$

and momentum

$$p = mv = \frac{m_0 v}{\sqrt{1 - v^2 / c^2}} \quad \dots(2)$$

Squaring equations (1) & (2)

$$E^2 = \frac{m_0^2 c^4}{1 - v^2 / c^2} \quad \dots(3)$$

$$p^2 = \frac{m_0^2 v^2}{1 - v^2 / c^2}$$

or

$$p^2 c^2 = \frac{m_0^2 v^2 c^2}{1 - v^2 / c^2} \quad \dots(4)$$

Subtracting equation (4) from equation (3),

$$E^2 - p^2 c^2 = \frac{m_0^2 c^4}{1 - v^2 / c^2} - \frac{m_0^2 v^2 c^2}{1 - v^2 / c^2}$$



$$E^2 - p^2 c^2 = \frac{m_0^2 c^4}{1 - v^2 / c^2} - \frac{m_0^2 v^2 c^2}{1 - v^2 / c^2}$$

$$= \frac{m_0^2 c^2 (c^2 - v^2)}{(1 - v^2 / c^2)}$$

$$= \frac{m_0^2 c^4 (1 - v^2 / c^2)}{(1 - v^2 / c^2)}$$

$$E^2 - p^2 c^2 = m_0^2 c^4$$

$$\therefore E^2 = p^2 c^2 + m_0^2 c^4$$

$$\therefore E = \sqrt{p^2 c^2 + m_0^2 c^4}$$

**This is an expression connecting relativistic energy and momentum.**

Since  $m_0$  and  $c$  are both Lorentz invariants,  $m_0 c^4$  is a constant and therefore, the quantity  $E^2 - p^2 c^2$  too is Lorentz invariant and its value remains unaltered in any inertial frame of reference.

### (iii) Mass, velocity, momentum and energy of particles of zero rest mass:-

We have just seen that the energy of a particle of rest mass  $m_0$  and momentum  $p$ , is given by

$$E = (p^2c^2 + m_0^2c^4)$$

If, therefore, we have a particle of rest mass zero, like a photon or a neutrino, clearly  $m_0 = 0$  and  $\therefore E = pc$ ,

**when  $P = E/C$  and since, as we have seen,  $P$  is also equal to  $Ev/c^2$ , [ $P=mv=> P= Ev/c^2$ ,  $E=mc^2$  ]**

we have

$$Ev/c^2 = E/c$$

$$\Rightarrow \mathbf{v = c}$$

**i.e., the velocity of particle having “zero rest mass” is always  $c$  (the speed of light) and is, therefore, invariant so its rest mass (zero), in all frames of reference.** and its energy is  $E = mc^2$ . Where  $m$  is the mass equivalent of energy  $E$ , equal to  $E/c^2$ .

Mass- Energy Conversion by HC Verma IIT kanpur  
[https://www.youtube.com/watch?v=\\_NRIYXvDkZk](https://www.youtube.com/watch?v=_NRIYXvDkZk)

What  $E=mc^2$  actually means? | Real life applications of  $E=mc^2$   
<https://www.youtube.com/watch?v=87QMpBHA1aw>

Proof of  $E=mc^2$  and Relativity of Mass  
<https://www.youtube.com/watch?v=wabnINynBGc>

Great interview:-Prof. Brian Greene Shows You How to Time Travel!  
<https://www.youtube.com/watch?v=DMK1MMjIm7I>

**Albert Einstein Biography**  
<https://www.youtube.com/watch?v=9qUa1YDWLmg>

The World in 2030" by Dr. Michio Kaku  
<https://www.youtube.com/watch?v=219YybX66MY>